ASIC DESIGN REPORT

CNN ACCELERATOR

NETLIST-TO-GDSII IMPLEMENTATION

ASIC physical implementation and optimization for a convolutional neural network accelerator.

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1. Project Overview

1.1. Project Objective

The objective of this project was to take the given CNN accelerator design through an ASIC physical implementation flow and obtain a valid RTL-to-GDSII implementation.

The work focused primarily on physical design and optimization, including floorplanning, placement, clock tree synthesis, routing, timing analysis, and physical verification. Design constraints and implementation parameters were also refined throughout the process when required by the observed results.

Rather than following a single fixed implementation flow, the project was developed iteratively. Timing violations, routing issues, congestion, design-rule violations, and other implementation problems were analyzed and addressed through multiple optimization attempts. The effectiveness of different approaches was evaluated by comparing implementation results across iterations.

The final objective was therefore not only to complete the RTL-to-GDSII flow, but to achieve a physically valid implementation with acceptable timing, design-rule compliance, and overall implementation quality.

1.2. Initial Design and Given Inputs

The post-synthesis report is shown in Figure 1. For this design, frequency was the primary objective, while area and power were treated as secondary priorities.

```
% source compile_power_sweep.tcl
freq_MHz        period_ns       slack_ns        area_um2        internal_mW     switch_mW       dynamic_mW      leakage_mW      total_mW        sram_instances
0.5             2000.0          997.80          1669687.562878  0.00159         0.161748        0.163338        5.13            5.3             42
1               1000.0          497.80          1669367.032881  0.003337        0.323875        0.327212        5.12            5.45            42
2               500.0           247.80          1669367.830880  0.006619        0.647704        0.654323        5.12            5.78            42
5               200.0           97.80           1669359.052881  0.016633        1.62            1.636633        5.12            6.76            42
10              100.0           47.80           1669374.480884  0.033416        3.24            3.273416        5.12            8.4             42
20              50.0            22.80           1669373.682884  0.067678        6.48            6.547678        5.12            11.7            42
50              20.0            7.80            1669375.012883  0.16947         16.2            16.36947        5.12            21.5            42
100             10.0            2.80            1669401.346882  0.337334        32.4            32.737334       5.12            37.9            42
150             6.66666666667   0.35            1669018.838894  0.342987        48.5            48.842987       5.12            54.0            42
200             5.0             0.00            1670357.084813  0.50105         64.7            65.20105        5.17            70.4            42
250             4.0             0.00            1674861.262476  0.549643        80.9            81.449643       5.31            86.7            42
300             3.33333333333   0.00            1686457.000322  0.857001        97.6            98.457001       5.56            104.0           42
350             2.85714285714   0.00            1690655.012118  1.09            116.0           117.09          5.69            122.0           42
385             2.5974025974    0.00            1692967.616037  1.15            127.0           128.15          5.79            134.0           42
400             2.5             -0.01           1693871.750027  1.24            132.0           133.24          5.81            139.0           42
425             2.35294117647   -0.09           1693724.386083  0.943641        140.0           140.943641      5.79            147.0           42
450             2.22222222222   -0.17           1694827.754143  1.05            149.0           150.05          5.87            156.0           42
475             2.10526315789   -0.21           1689821.633889  1.55            157.0           158.55          5.58            164.0           42
500             2.0             -0.28           1689007.673939  1.16            165.0           166.16          5.59            172.0           42
550             1.81818181818   -0.36           1690152.538001  1.4             182.0           183.4           5.64            189.0           42
600             1.66666666667   -0.46           1690354.432044  1.22            198.0           199.22          5.67            205.0           42
650             1.53846153846   -0.51           1690777.904007  1.3             215.0           216.3           5.68            222.0           42
700             1.42857142857   -0.57           1695458.971825  1.41            231.0           232.41          5.79            238.0           42
750             1.33333333333   -0.62           1696258.567840  1.57            248.0           249.57          5.83            255.0           42
800             1.25            -0.64           1698709.757824  1.7             264.0           265.7           5.91            272.0           42
%
```

Figure 1: Synthesis Report Table

The synthesis summary indicates that the design can meet timing at 385 MHz in the best case. Because the implementation had not yet progressed to physical design, that performance target was considered optimistic. A more conservative and achievable operating point of 250 MHz was selected for the ASIC flow.

1.3. Target Technology and Design Flow

The physical implementation was carried out from netlist to GDSII in Cadence Innovus using the gsclib045 (N45) library. This library provides standard cells, SRAM macros, and physical

cells appropriate for a high-speed accelerator implementation.

The provided technology files include .lib and .lef data for the FF (fast), SS (slow), and TP (typical) corners, as well as captbl files for RC extraction. The physical cell library also contains WELLTAP and FILLER cells, though ENDCAP cells were not available.

2. Design Challenges

Meeting timing constraints was the most significant challenge in this project. The inclusion of 42 SRAM macros required a careful floorplan to balance memory placement, cell placement, and routing resources.

Power planning was also important: the power grid had to be designed to prevent IR drop and minimize leakage while maintaining adequate distribution to the entire design. I/O pin planning was deferred until after timing closure had been achieved, in order to focus first on the core physical implementation.

3. Optimization Experiments

3.1. Optimization Strategy

The implementation flow was developed iteratively. Each experiment tested a different physical strategy, and the results were evaluated at placement, clock-tree synthesis (CTS), and routing stages.

3.2. Experiment 1

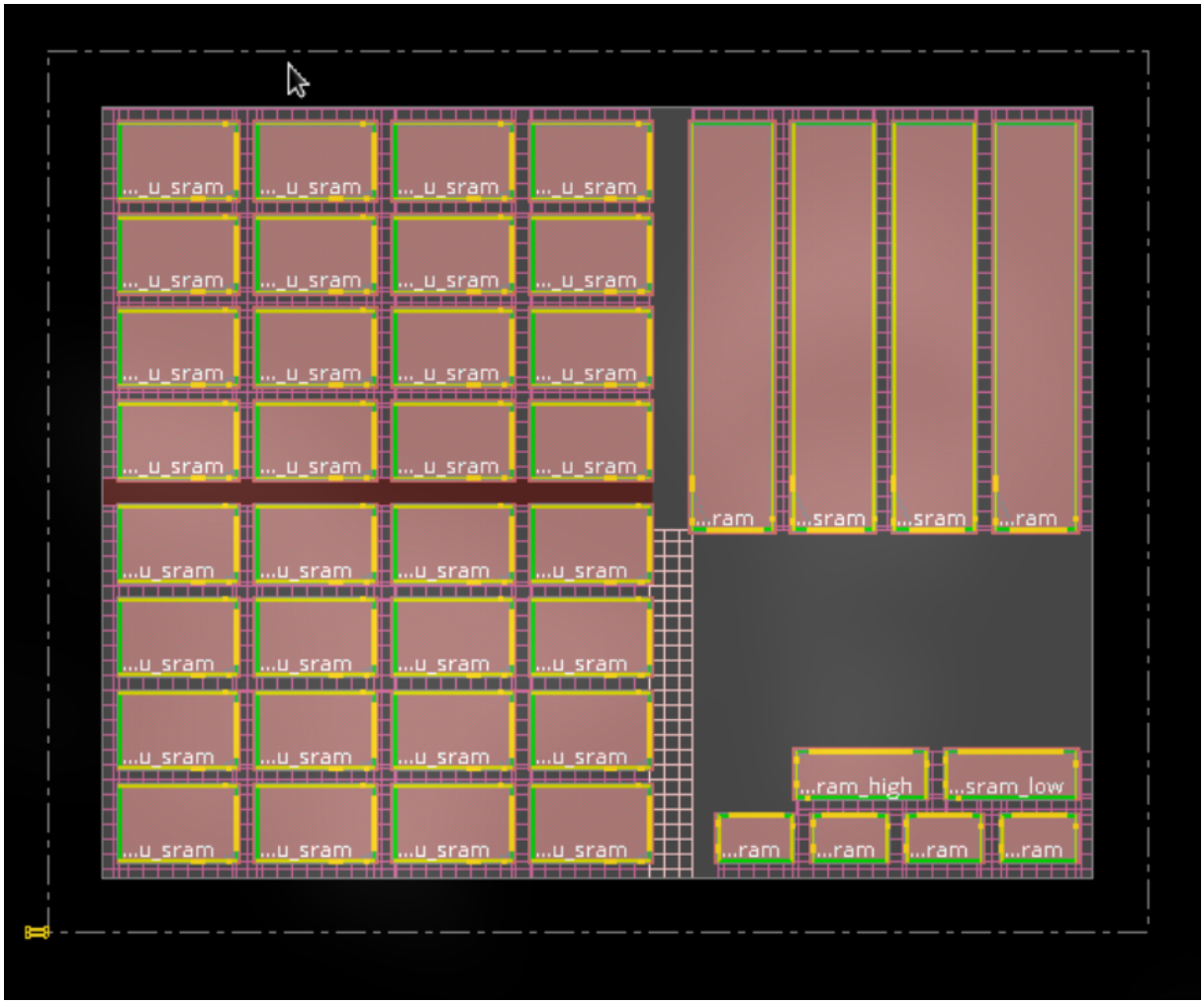


Figure 2: Initial floorplan

The first floorplan exposed a key issue: the SRAM macros were rotated 90 degrees and ultimately violated DRC rules. Although the layout left space for standard cells, the resulting physical organization placed the core logic far from the memory macros in the top-left corner.

This version was still useful for establishing the power-distribution strategy. The power ring was built on Metal 9 and Metal 10, with rails and stripes on the same layers and vias dropped to Metal 1. Because only a single stripe layer could carry power, that layer was made denser to improve distribution.

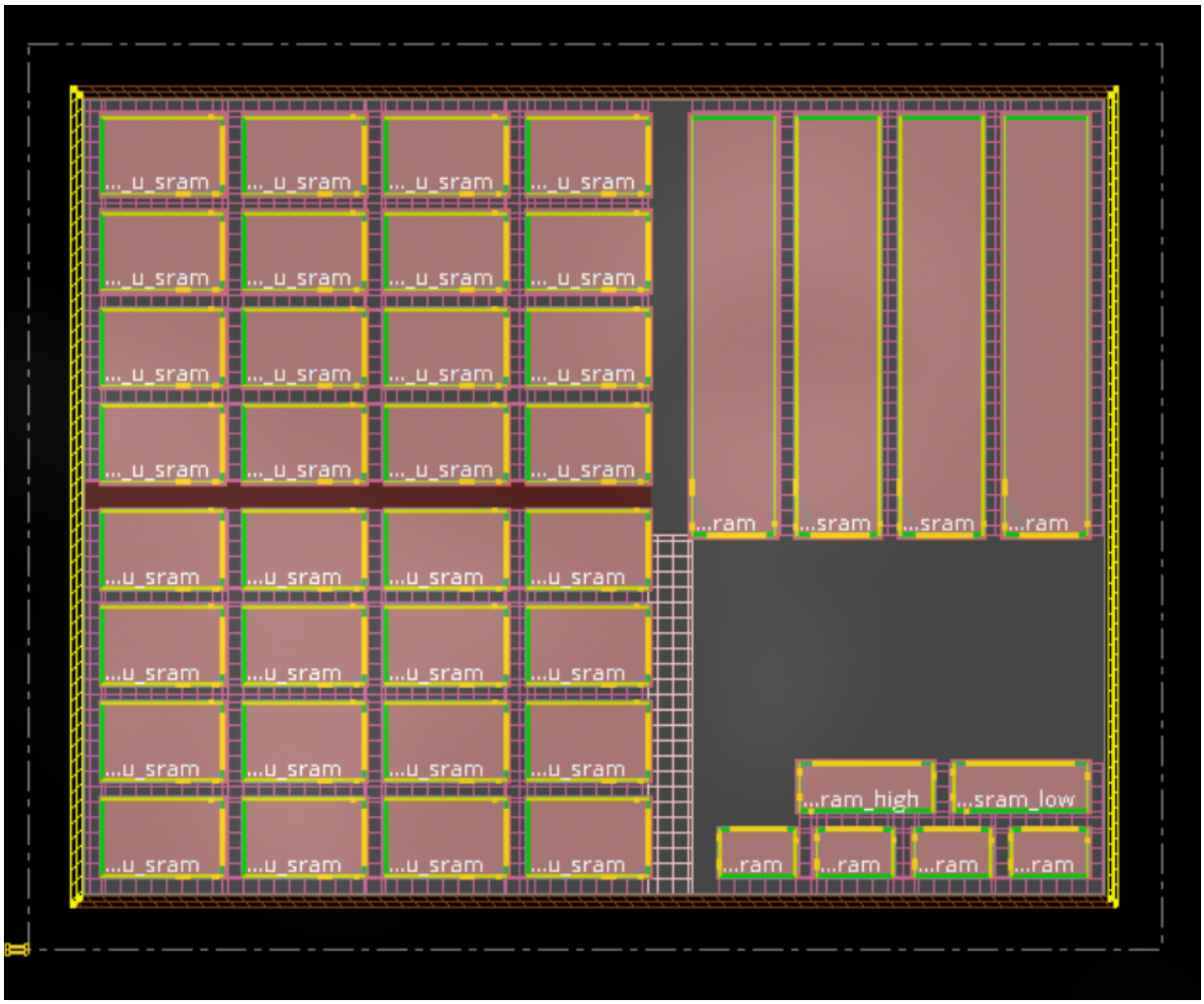


Figure 3: Build ring

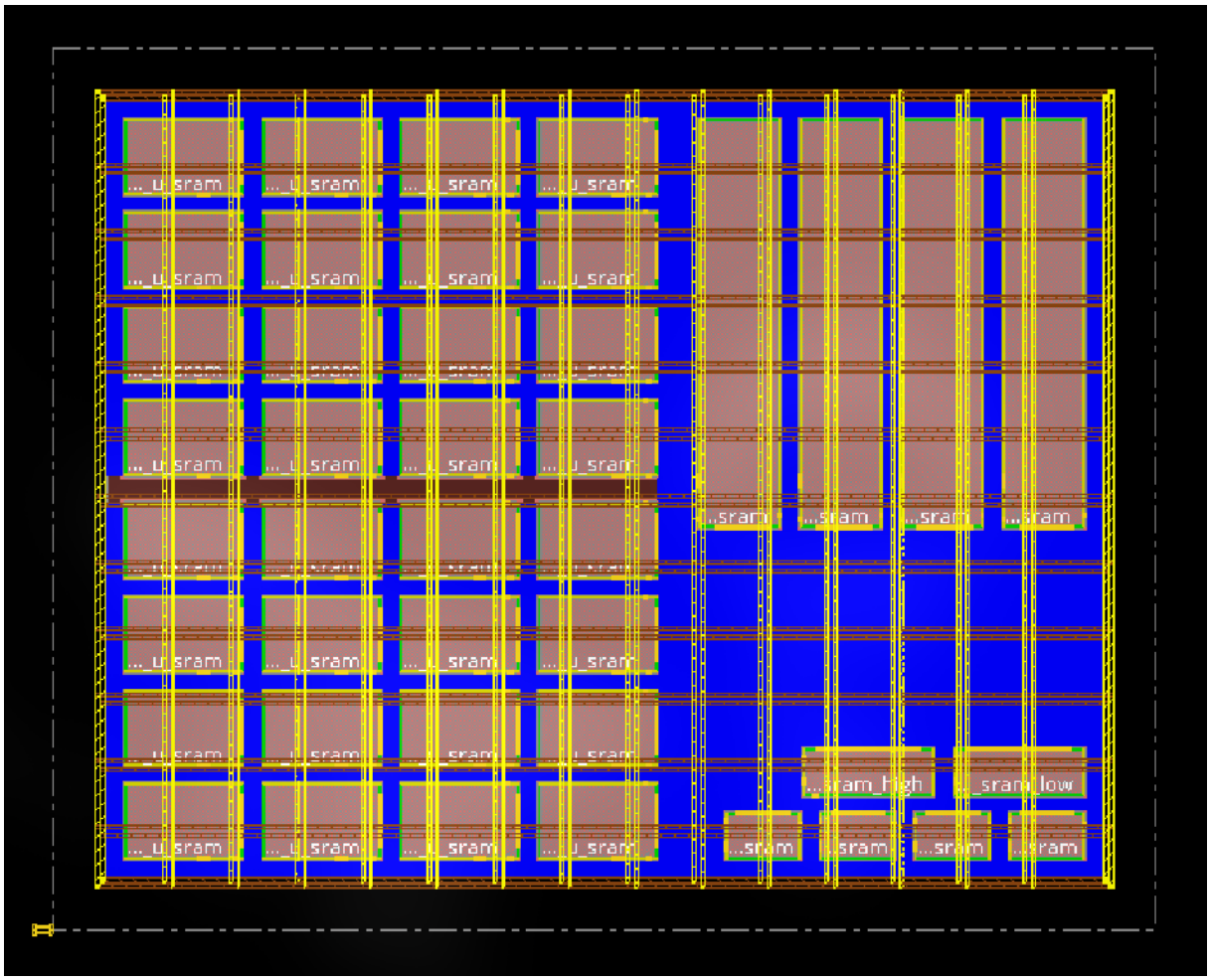


Figure 4: Build stripes and rail

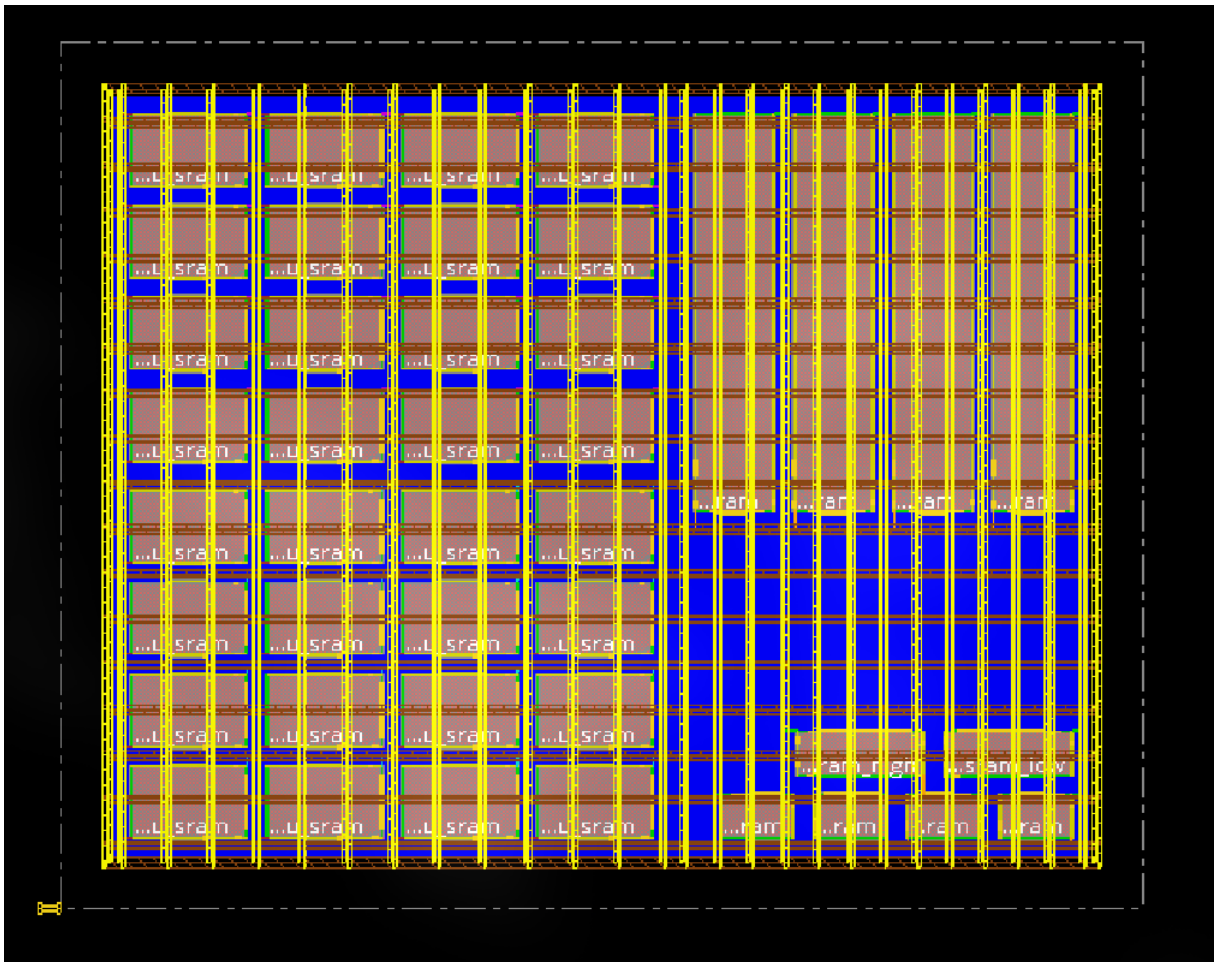


Figure 5: Rebuild stripe

A preliminary check indicated that the power network was substantially in place. The red markers in Figure 5 show the via1 connections, confirming completion of the PG build.

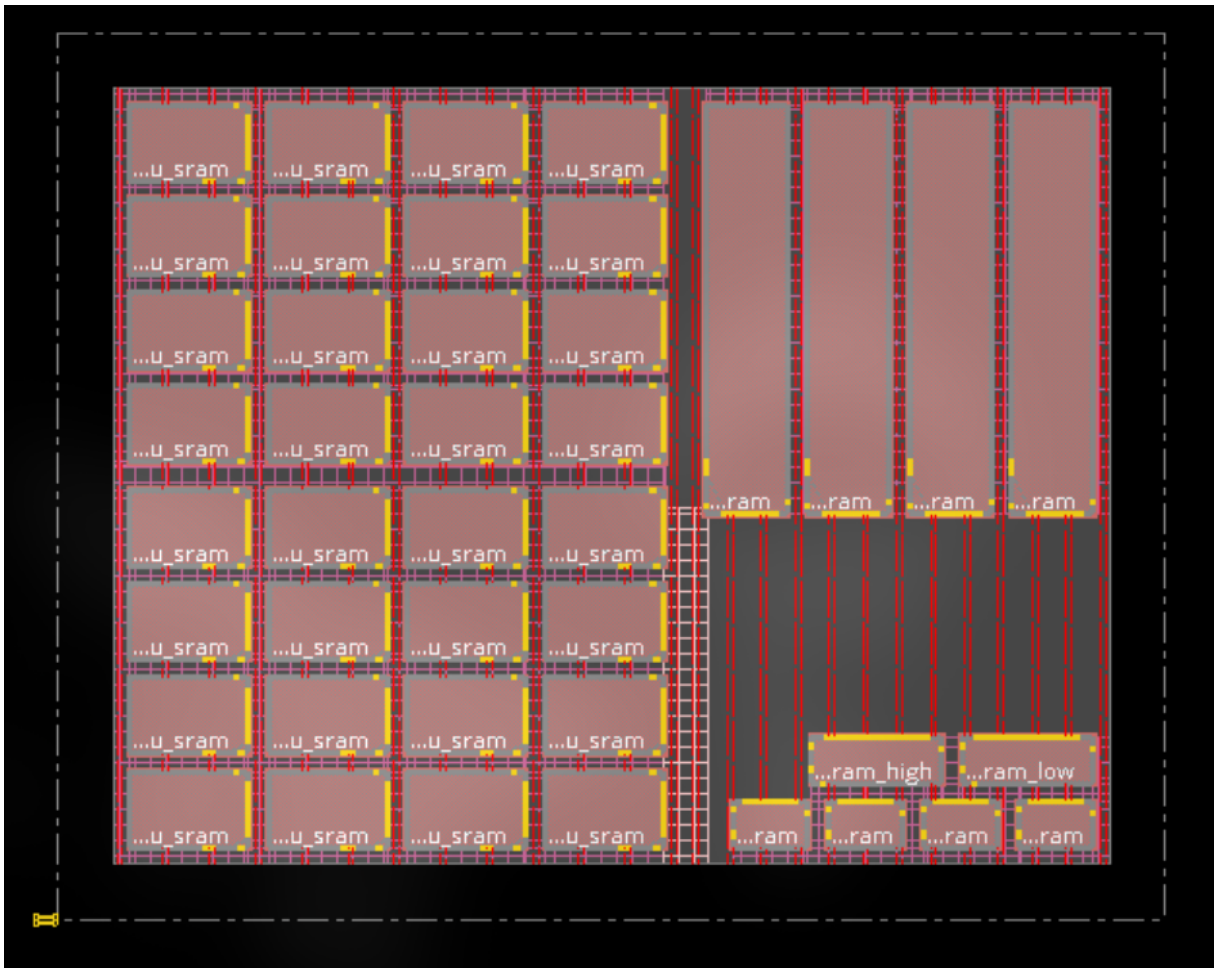


Figure 6: Check via

```
Total Power
----------------------------------------------------------------------------------------
Total Internal Power:      104.82375812            74.3055%
Total Switching Power:      31.36449941            22.2331%
Total Leakage Power:         4.88306040             3.4614%
Total Power:               141.07131793
----------------------------------------------------------------------------------------


Group                          Internal   Switching     Leakage       Total  Percentage
                               Power      Power         Power         Power  (%)
----------------------------------------------------------------------------------------
Sequential                        19.31       1.919       1.041       22.27       15.78
Macro                             55.35     0.02544      0.8514       56.22       39.85
IO                                    0           0           0           0           0
Combinational                     29.95       29.42       2.984       62.35        44.2
Clock (Combinational)                 0           0           0           0           0
Clock (Sequential)               0.2252           0    0.006207      0.2314       0.164
----------------------------------------------------------------------------------------
Total                             104.8       31.36       4.883       141.1         100
----------------------------------------------------------------------------------------


Rail                 Voltage   Internal   Switching     Leakage       Total  Percentage
                               Power      Power         Power         Power  (%)
----------------------------------------------------------------------------------------
VDD                     0.95      104.8       31.36       4.883       141.1         100


----------------------------------------------------------------------------------------
*       Power Distribution Summary:
*               Highest Average Power: u_weight_memory/u_bank0/gen_asic_memory_gen_sram_16x4096_u_sram (SRAM_16x4096_lrw):            4.34
*               Highest Leakage Power: u_weight_memory/u_bank0/gen_asic_memory_gen_sram_16x4096_u_sram (SRAM_16x4096_lrw):         0.06777
*          Total Cap:      7.12685e-10 F
*          Total instances in design: 173297
*          Total instances in design with no power:   152
*          Total instances in design with no activity:   152
*          Total Fillers and Decap:     0
----------------------------------------------------------------------------------------
```

Figure 7: Power report

At placement, the estimated total power was 141.07 mW at 0.95 V. Internal power accounted for 104.82 mW (74.31%), switching power was 31.36 mW (22.23%), and leakage power was 4.88 mW (3.46%). Macro logic contributed 39.85% and combinational logic contributed 44.20% of the total.

This estimate was preliminary because the design had not yet reached post-route extraction or final timing analysis. The timing performance of this version was insufficient; the 250 MHz target could not be reached, so the design was revised for the next experiment.

3.3. Experiment 2

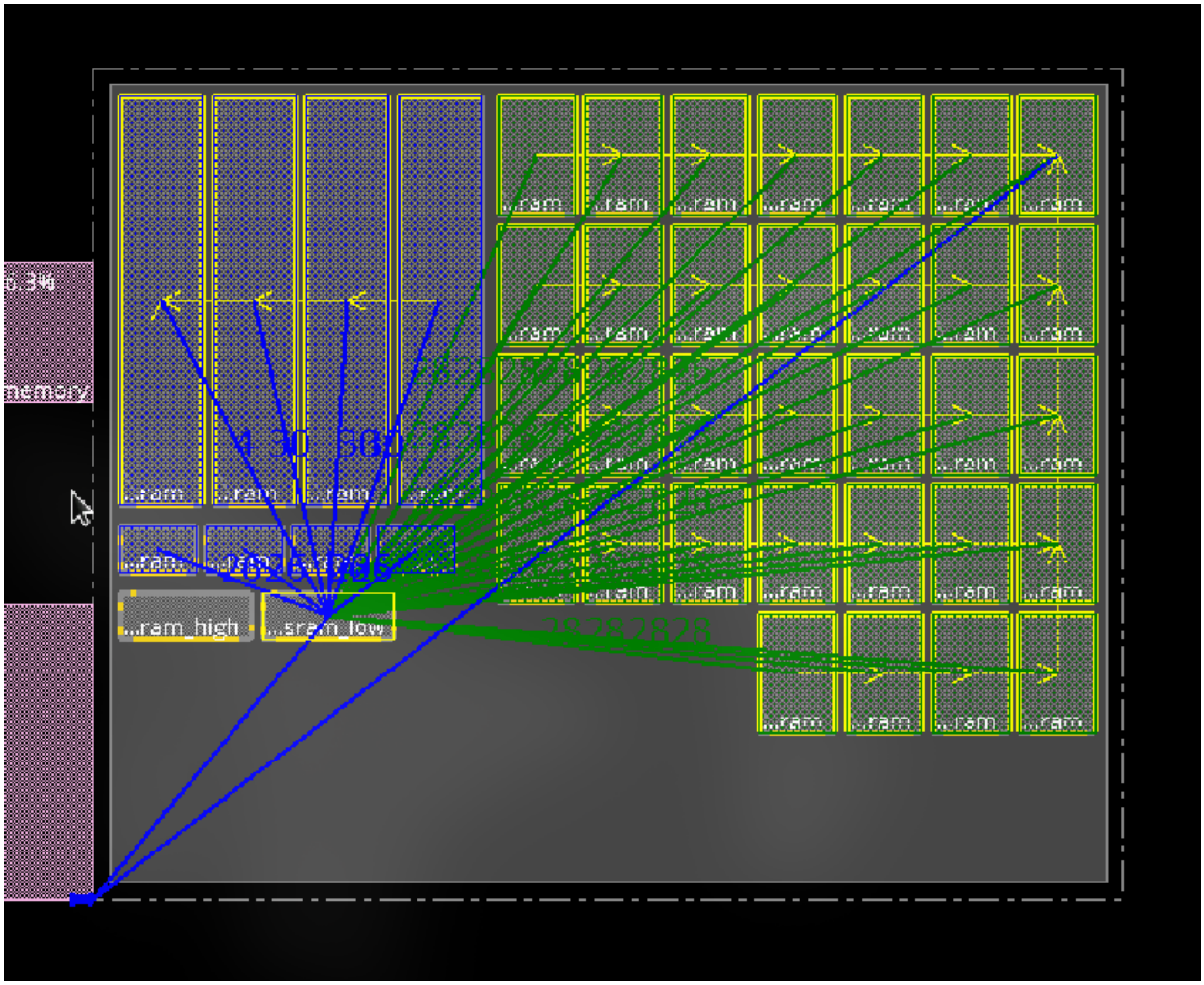


Figure 8: Floorplan v2

This second arrangement eliminated the SRAM DRC issues and provided more space for standard-cell placement.

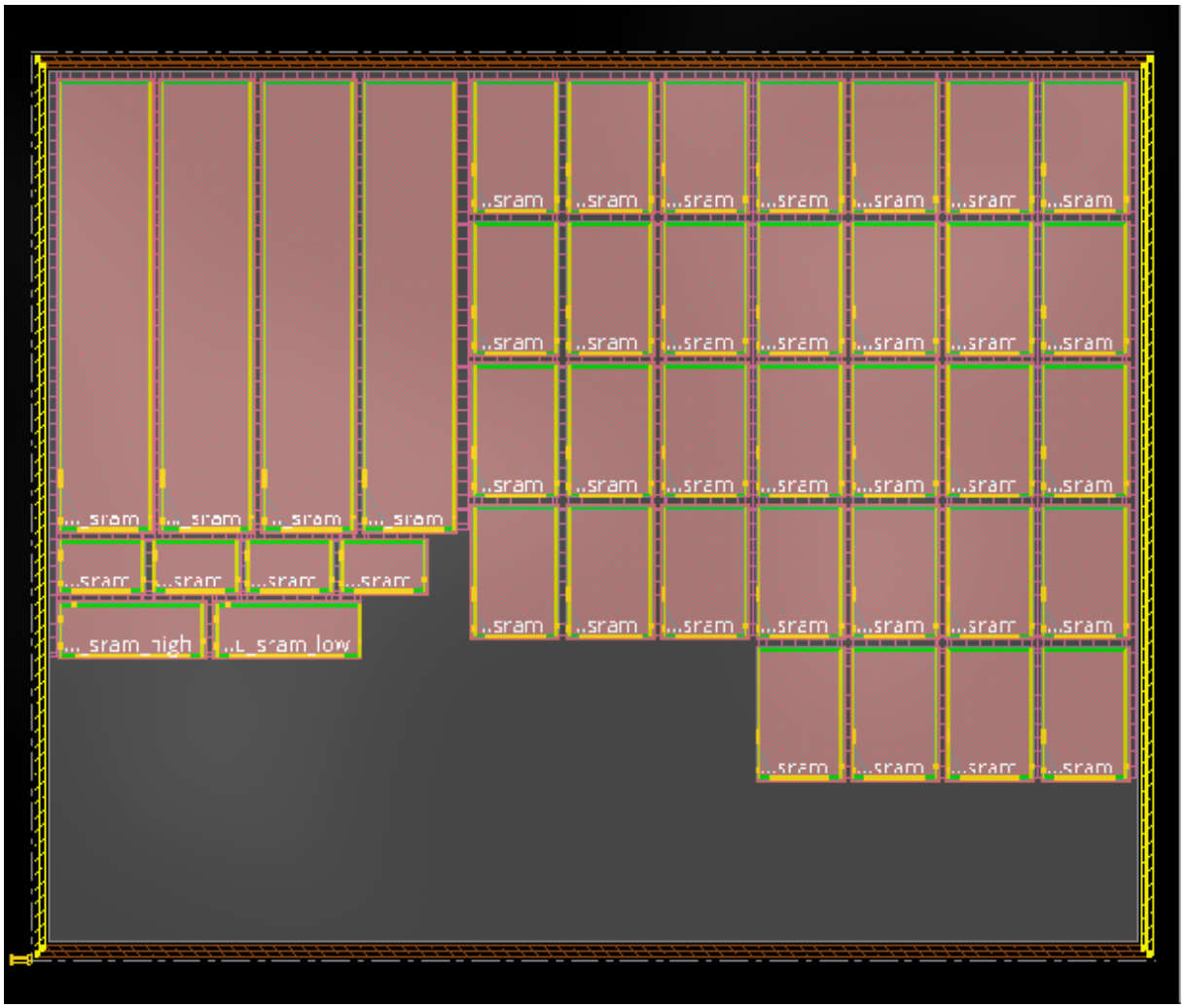


Figure 9: Build ring

The previous power-grid script was reused for this case.

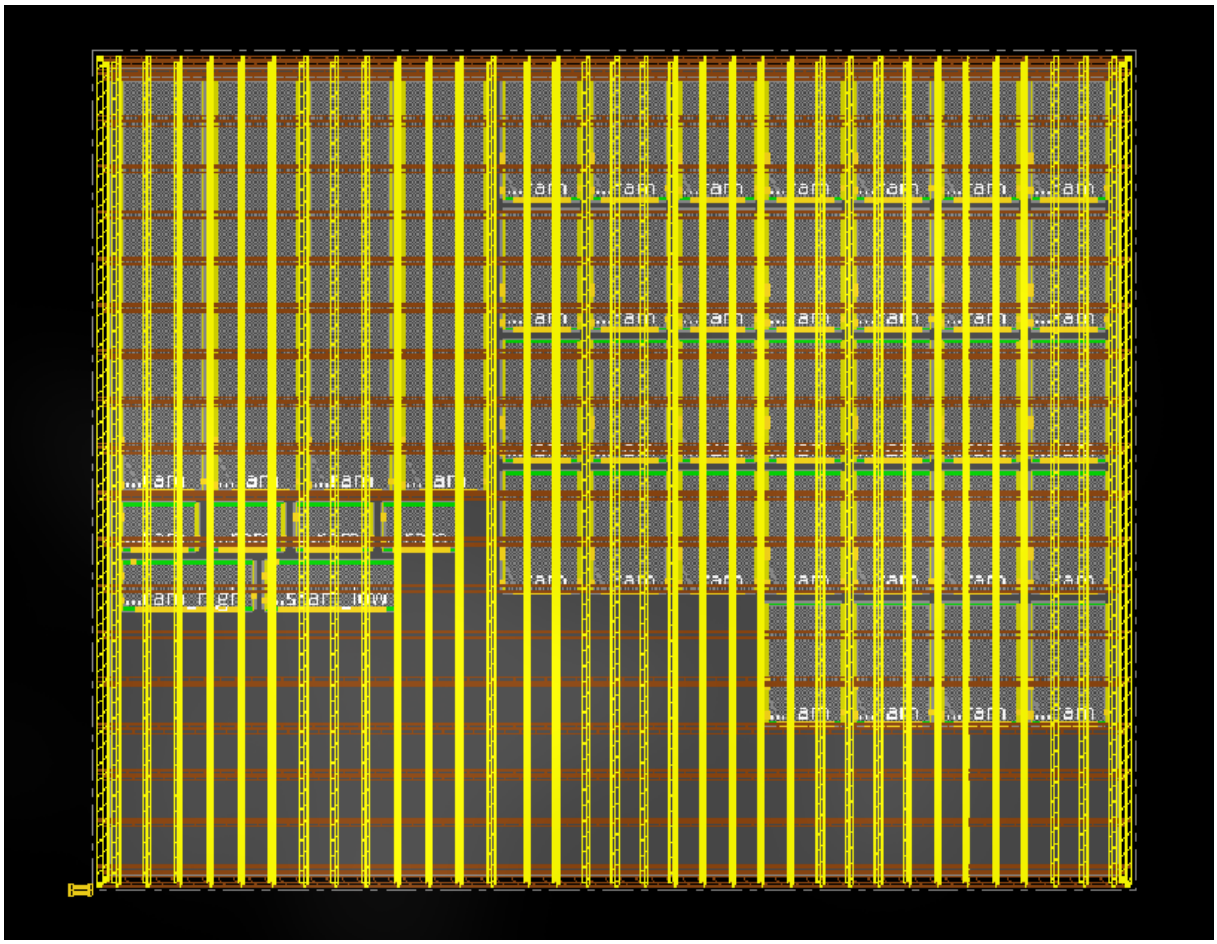


Figure 10: Build stripe & rail

The I/O assignment was manually refined instead of relying entirely on automatic tool placement.

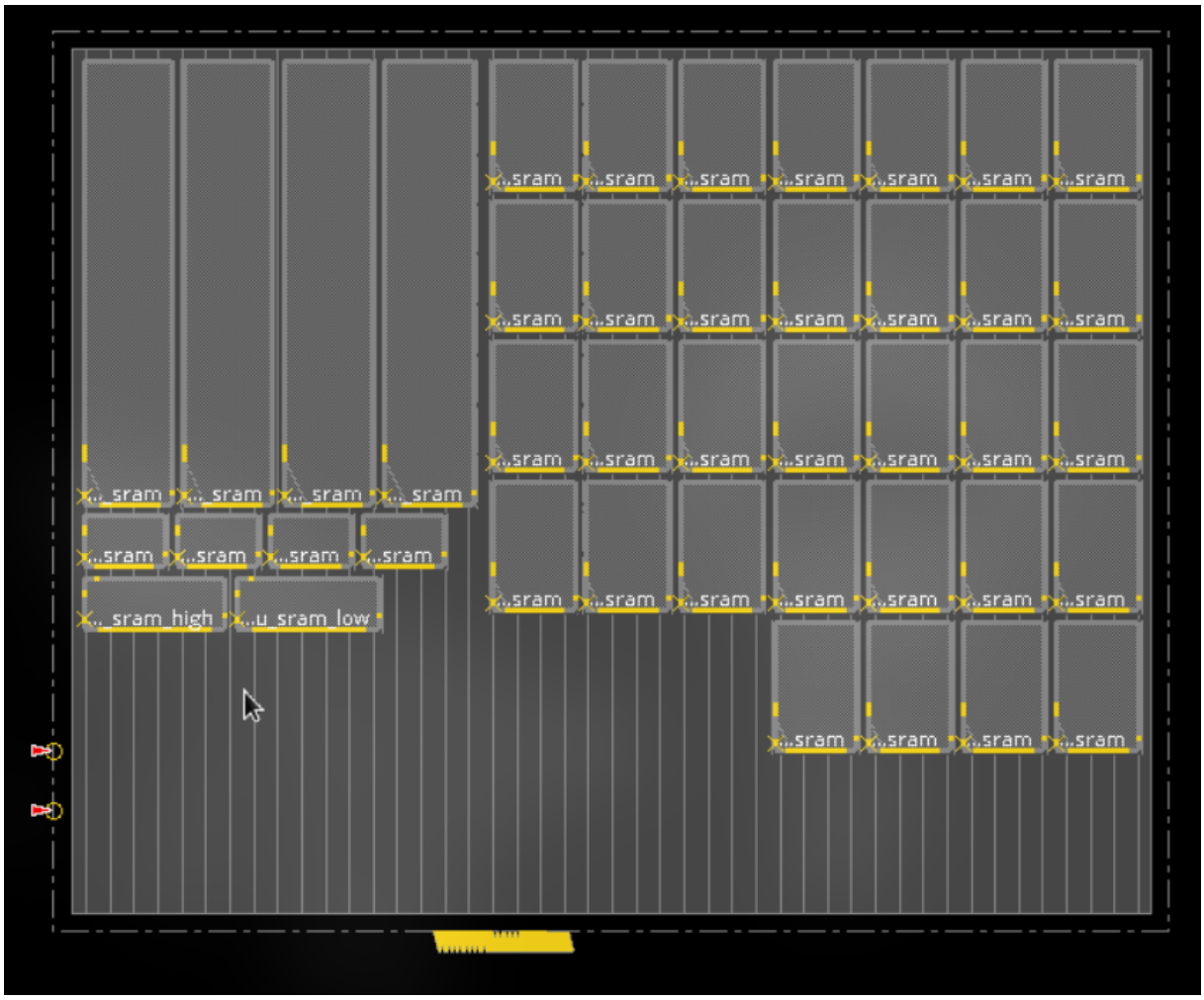


Figure 11: Edit pin

After placement optimization, the WNS for reg-to-reg paths improved to -0.005 ns with 51 violating paths, indicating that 250 MHz was becoming a realistic target.

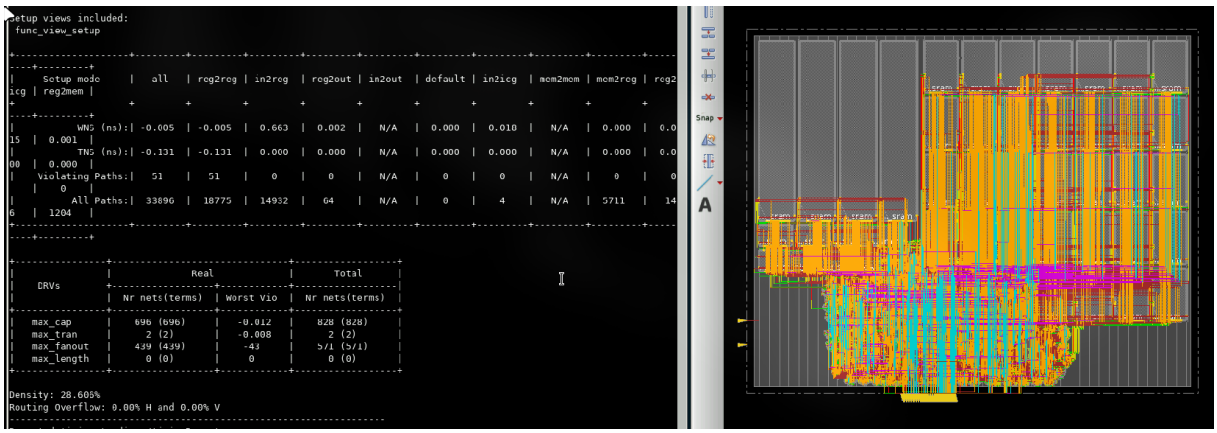


Figure 12: preCTS

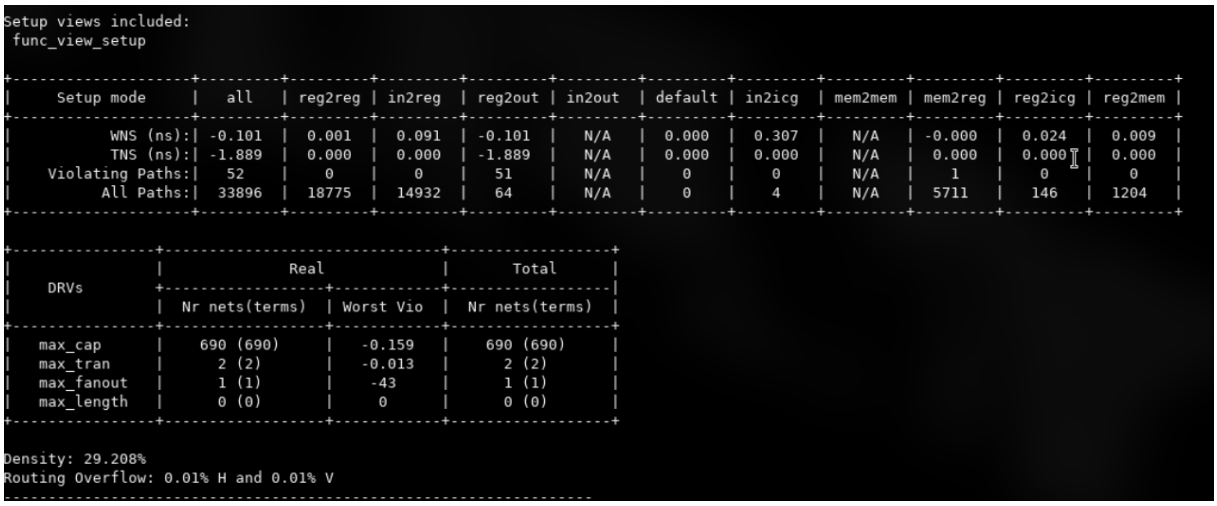


Figure 13: postCTS

During CTS—after optimization and the application of useful skew—reg-to-reg timing was fully met, but a violation remained on the reg-to-out path with WNS -0.101. Although the tool labeled this path as reg-to-out, the critical path actually originated from an SRAM output. Memory-to-output paths are difficult to improve using standard-cell optimization methods.

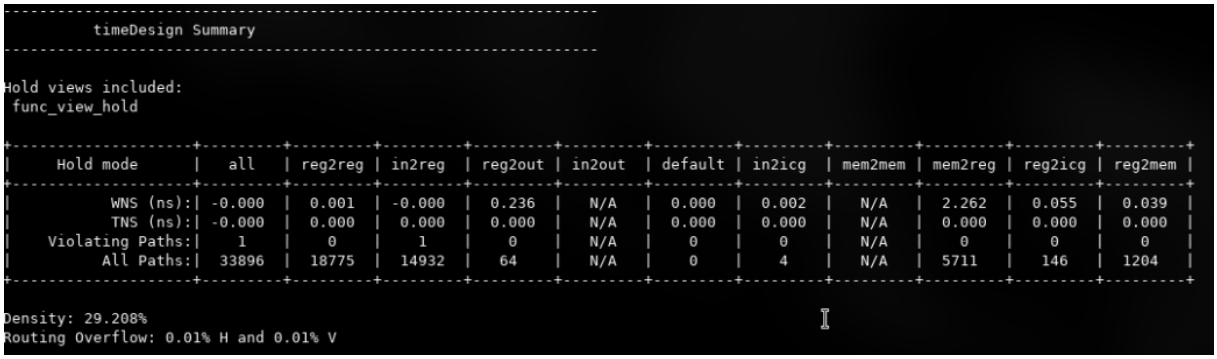


Figure 14: postCTS_hold

Hold timing was acceptable, with only one path violation remaining.

At the routing stage, the design reached a level of complexity that could not be fully resolved by the available optimization options. Additional implementation experiments were run in parallel, including a 300 MHz case, which suggested that higher frequency might still be feasible with further refinement.

3.4. Experiment 3

This experiment revisited the floorplan, power grid, and I/O placement with a stronger emphasis on architectural layering.

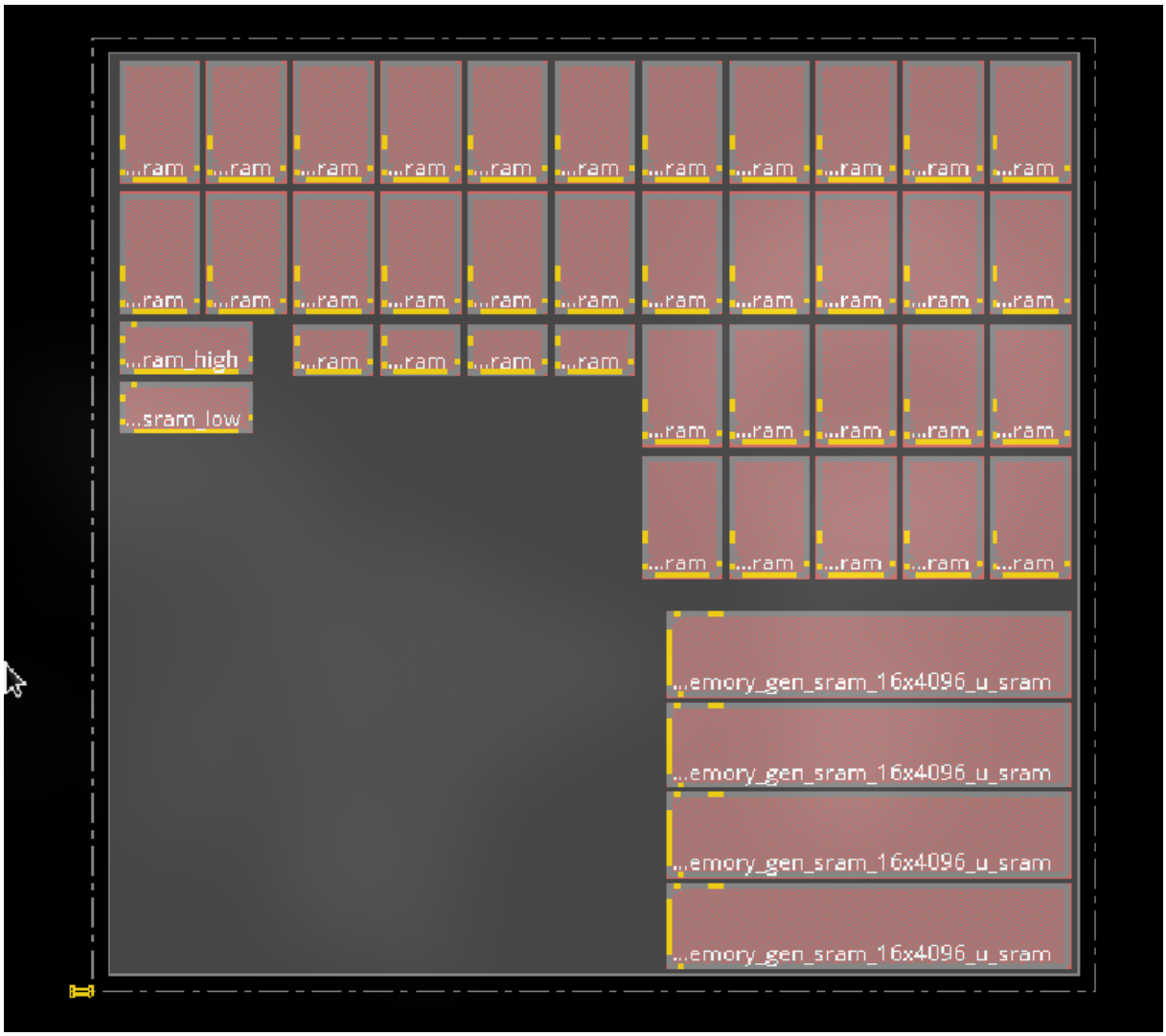


Figure 15: Floorplan_v3

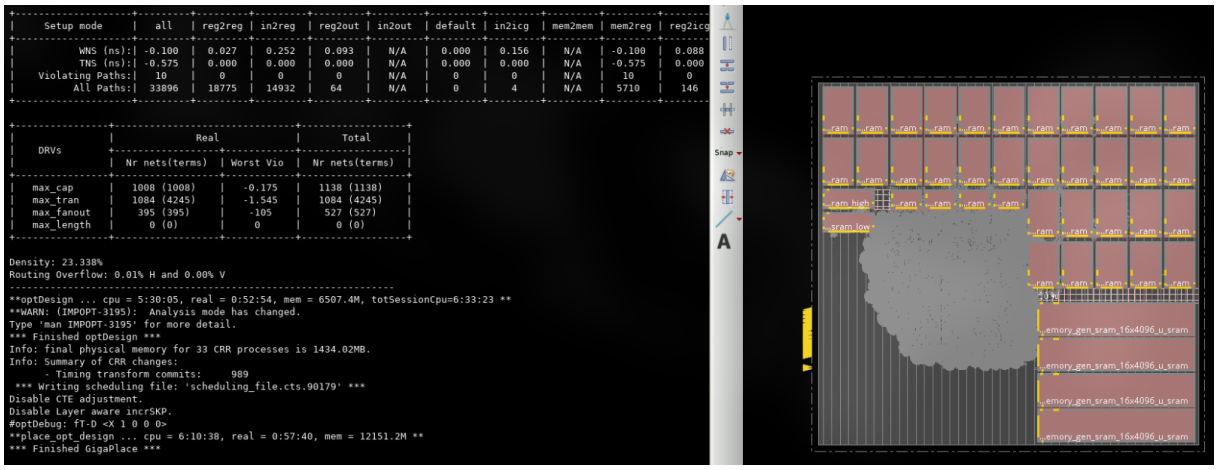


Figure 16: preCTS

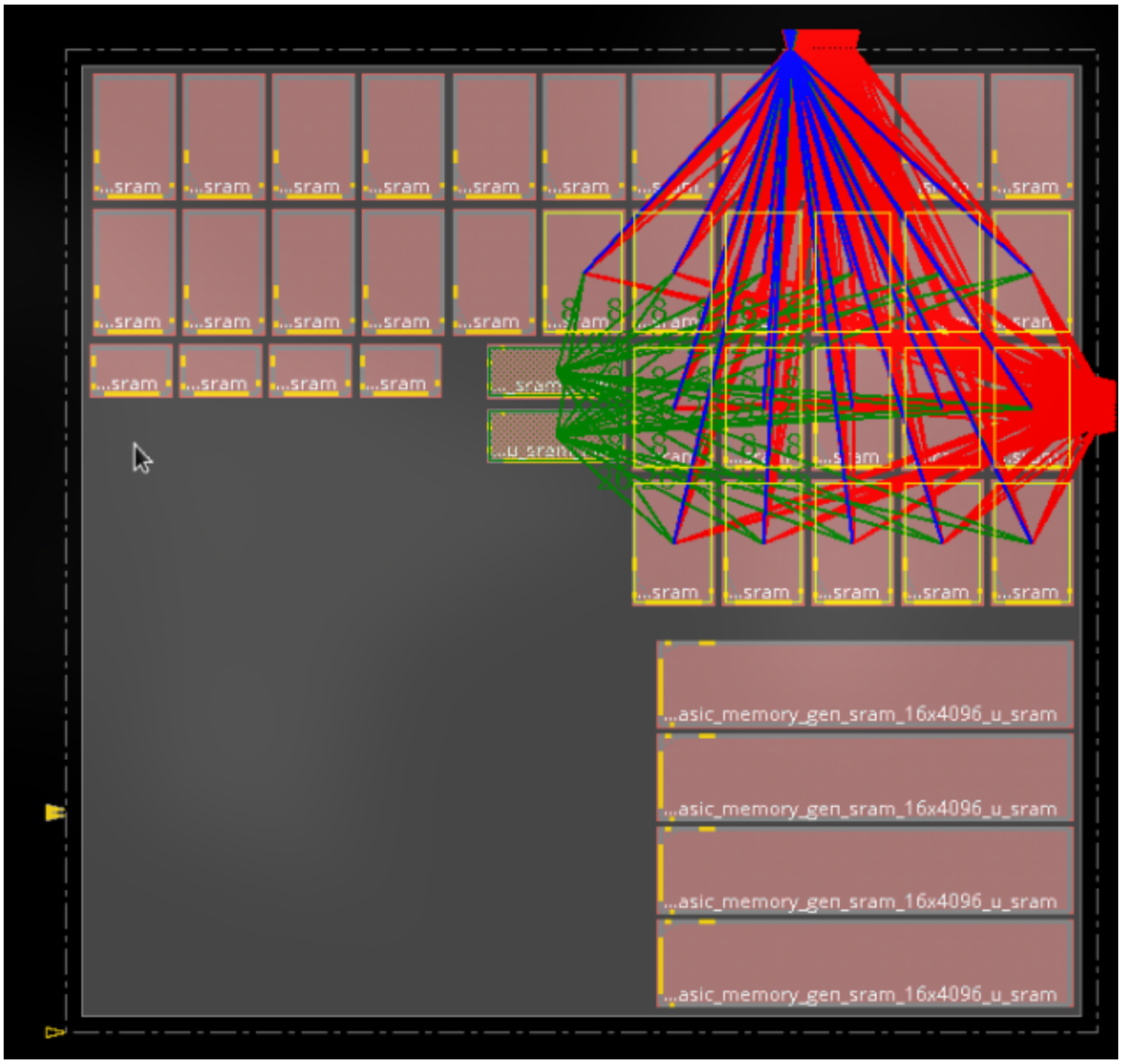


Figure 17: adjust u_instruction_memory

The ‘u_instruction_memory‘ block was moved closer to the center of the die, and the I/O pins were relocated toward the top. These changes reduced the critical distances between SRAM and core logic.

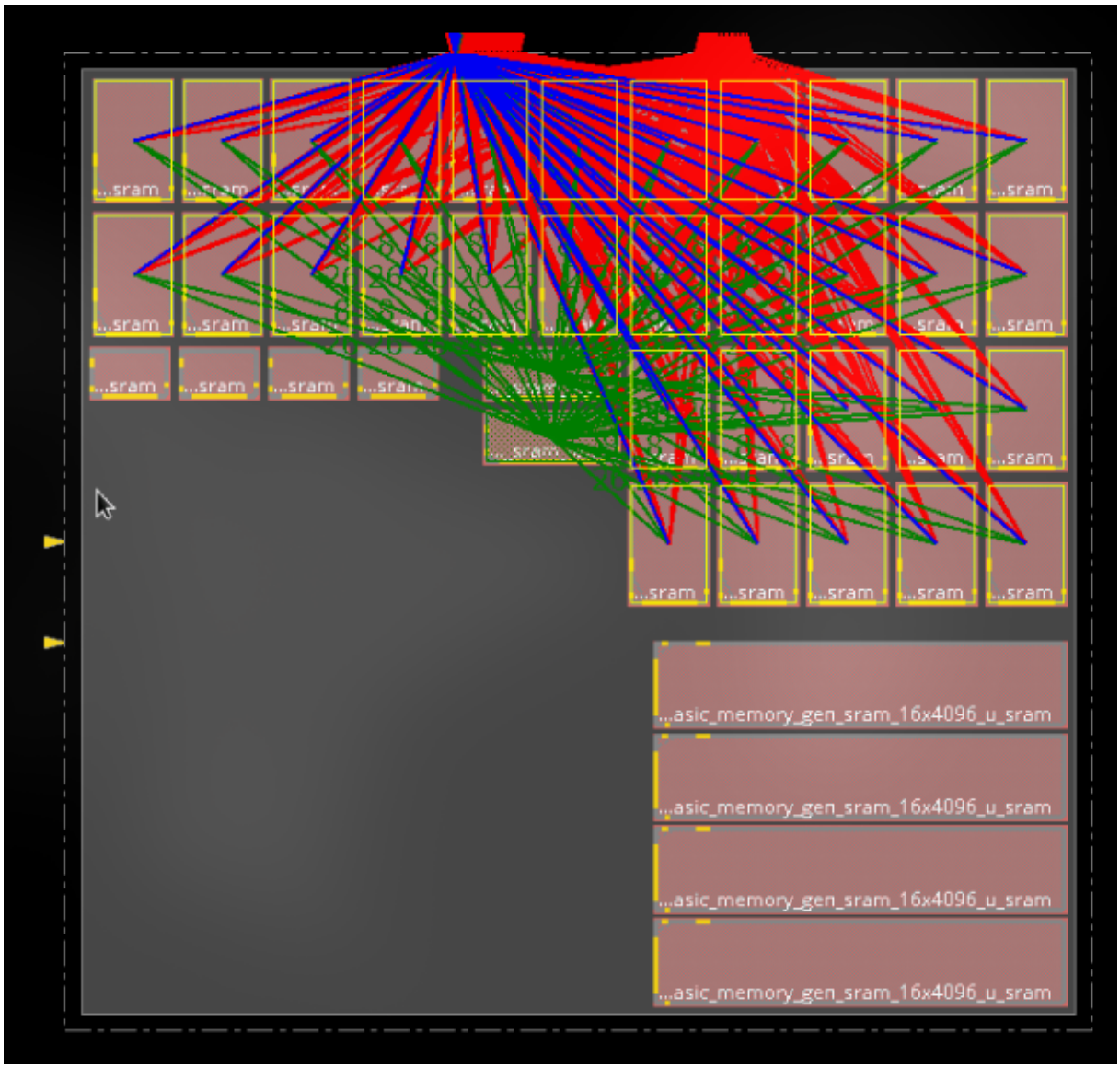


Figure 18: u_instruction_memory flight line

Additional block placement constraints were applied to the core. The table below lists the affected blocks and their assigned constraint types.

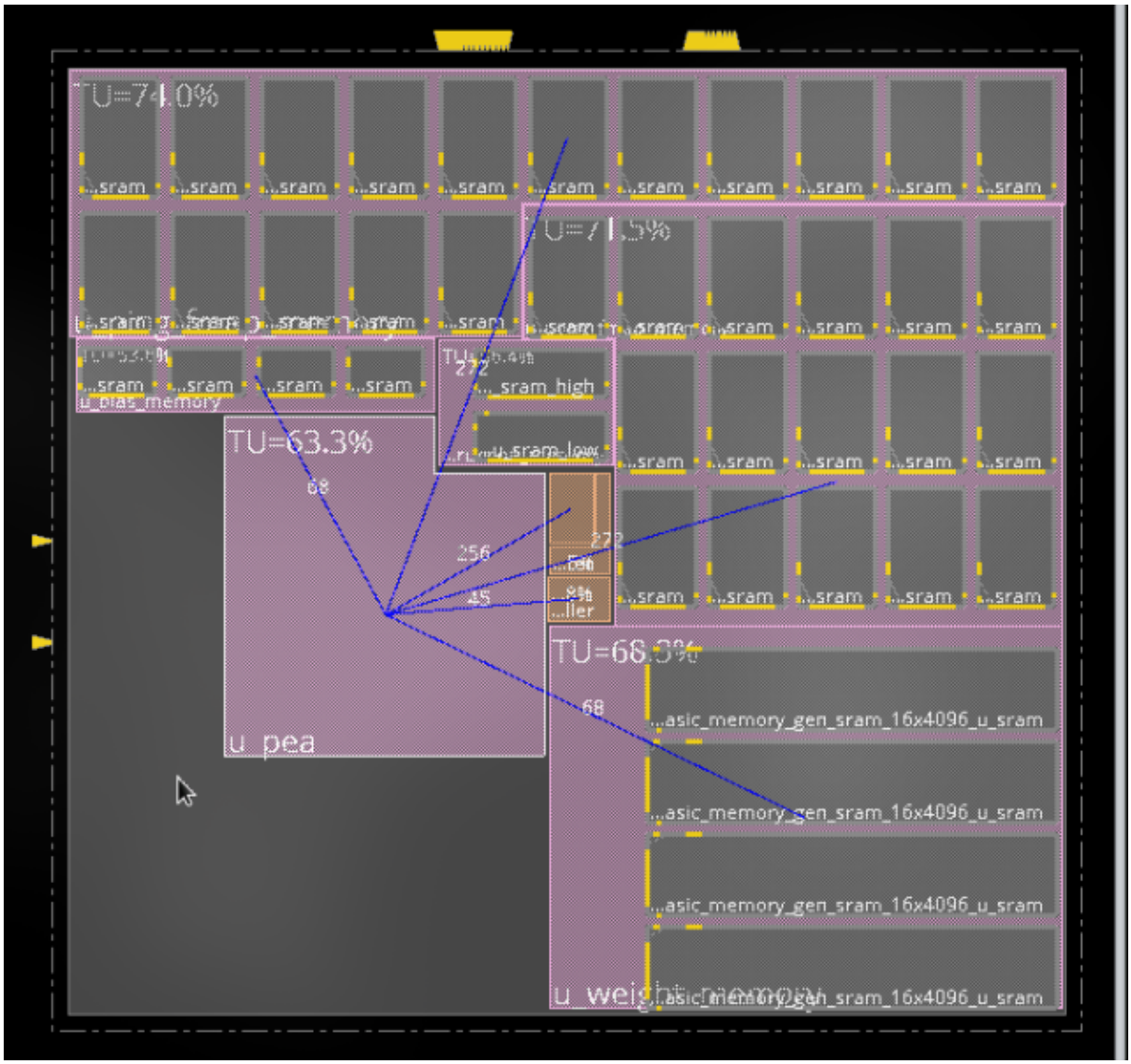


Figure 19: Placement constraints

Table 1: Initial Physical Constraints

Instance	Constraint Type	Color
u_global_arbiter	Region	Orange
u_controller	Region	Orange
u_ifm_arbiter	Region	Orange
u_line_buffer	Region	Orange
u_bias_memory	Guide	Purple
u_weight_memory	Guide	Purple
u_instruction_memory	Guide	Purple
u_pong_fmap_memory	Guide	Purple
u_ping_fmap_memory	Guide	Purple

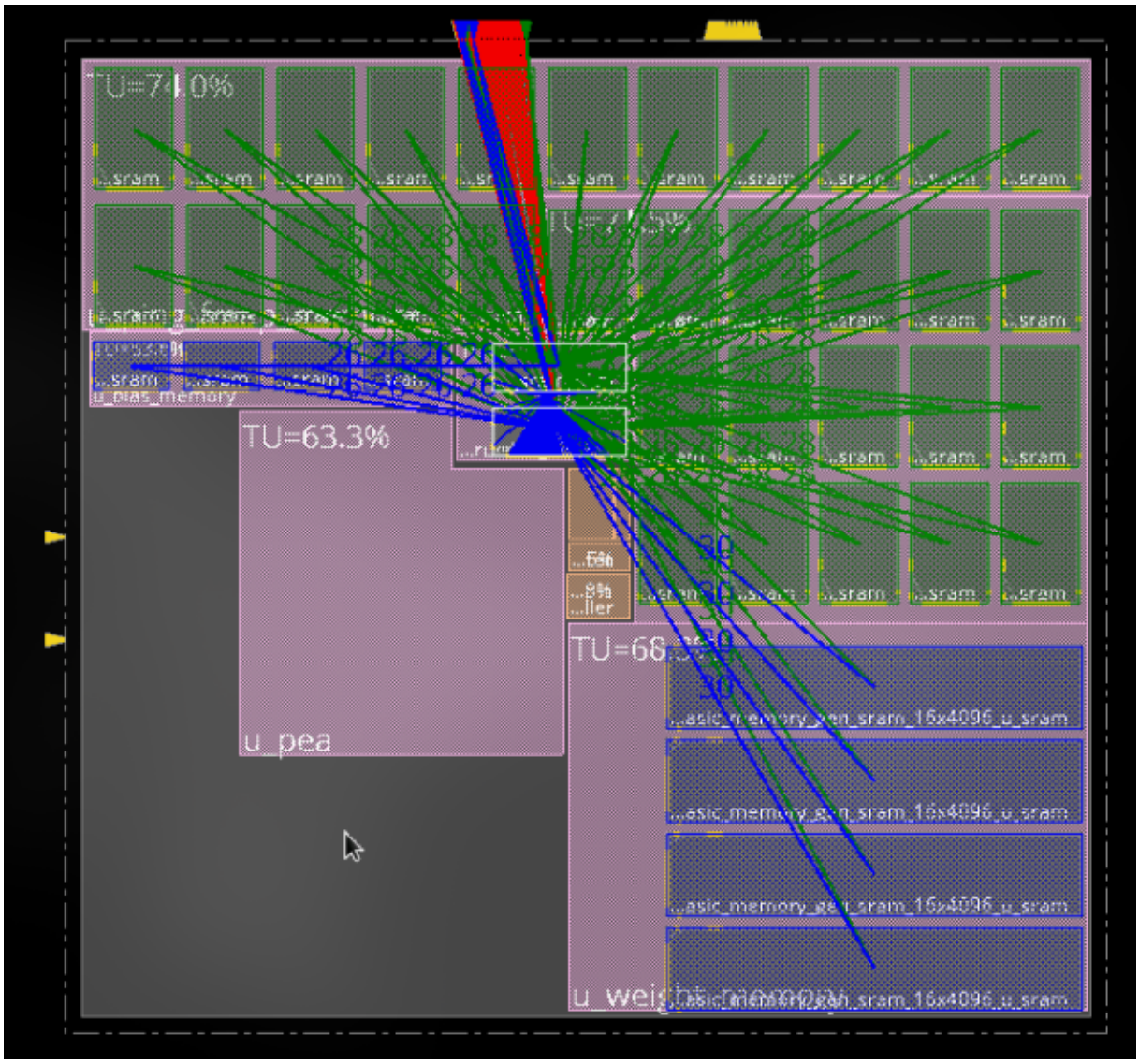


Figure 20: Overview after adding placement constraints

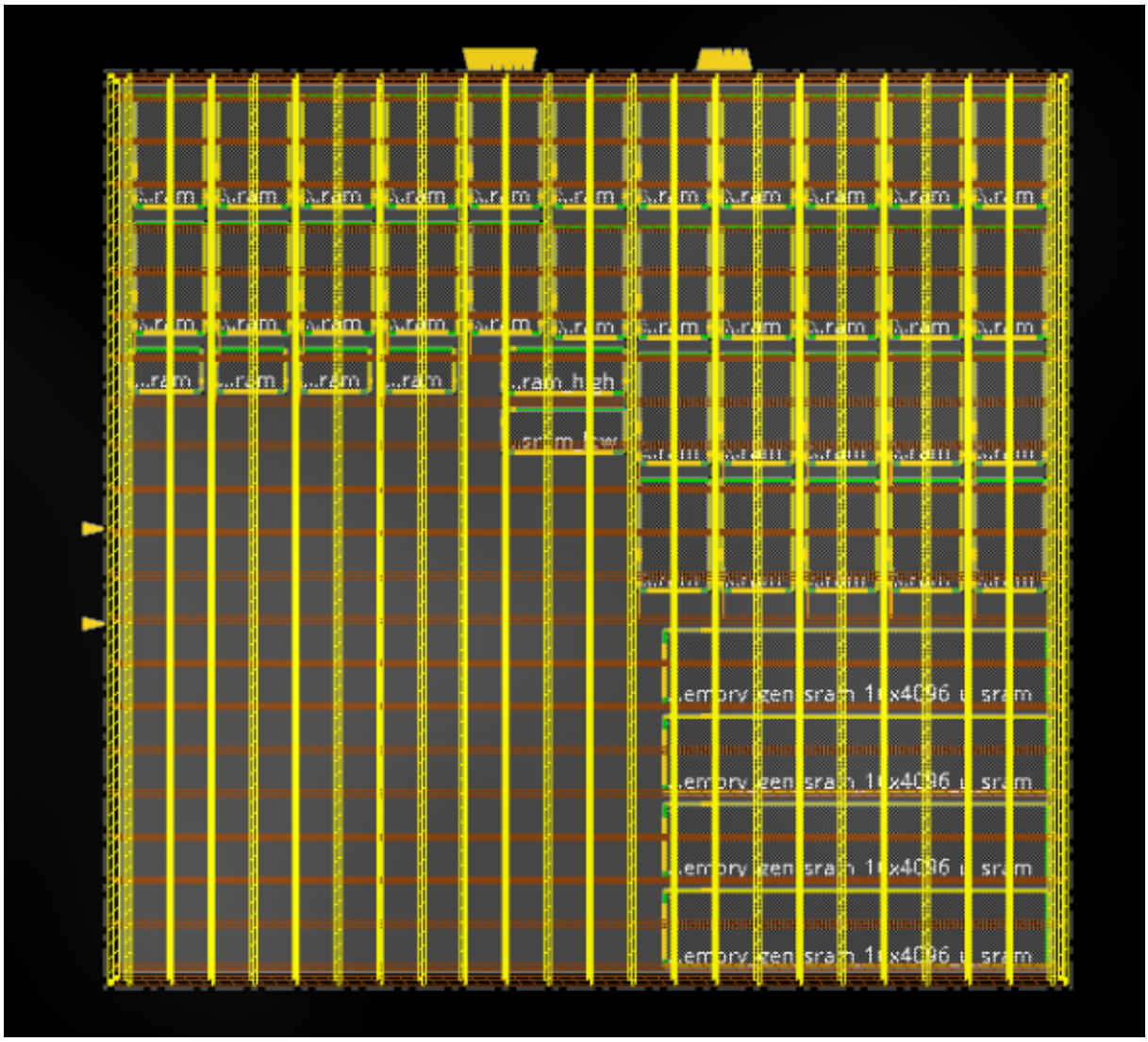


Figure 21: Build PG

```
Setup views included:
 func_view_setup

+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | in2reg  | reg2out | in2out  | default | in2icg  | mem2mem | mem2reg | reg2icg | reg2mem |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):| -0.092  | -0.031  |  0.182  |  0.067  |   N/A   |  0.000  |  0.076  |   N/A   | -0.092  |  0.060  |  0.060  |
|           TNS (ns):| -1.355  | -0.653  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |   N/A   | -0.702  |  0.000  |  0.000  |
|    Violating Paths:|   67    |   54    |    0    |    0    |   N/A   |    0    |    0    |   N/A   |   13    |    0    |    0    |
|         All Paths:|  33896  |  18775  |  14932  |   64    |   N/A   |    0    |    4    |   N/A   |  5710   |   146   |  1204   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+

+---------------+-----------------------------+-----------------+
|               |            Real             |      Total      |
|     DRVs      +-----------------+-----------+                 |
|               |  Nr nets(terms) | Worst Vio |  Nr nets(terms) |
+---------------+-----------------+-----------+-----------------+
|   max_cap     |    976 (976)    |   -0.179  |   1106 (1106)   |
|   max_tran    |   1022 (3291)   |   -1.480  |   1022 (3291)   |
|   max_fanout  |    555 (555)    |    -51    |    688 (688)    |
|   max_length  |      0 (0)      |     0     |      0 (0)      |
+---------------+-----------------+-----------+-----------------+
```

Figure 22: preCTS_v3

Timing results improved significantly after these changes.

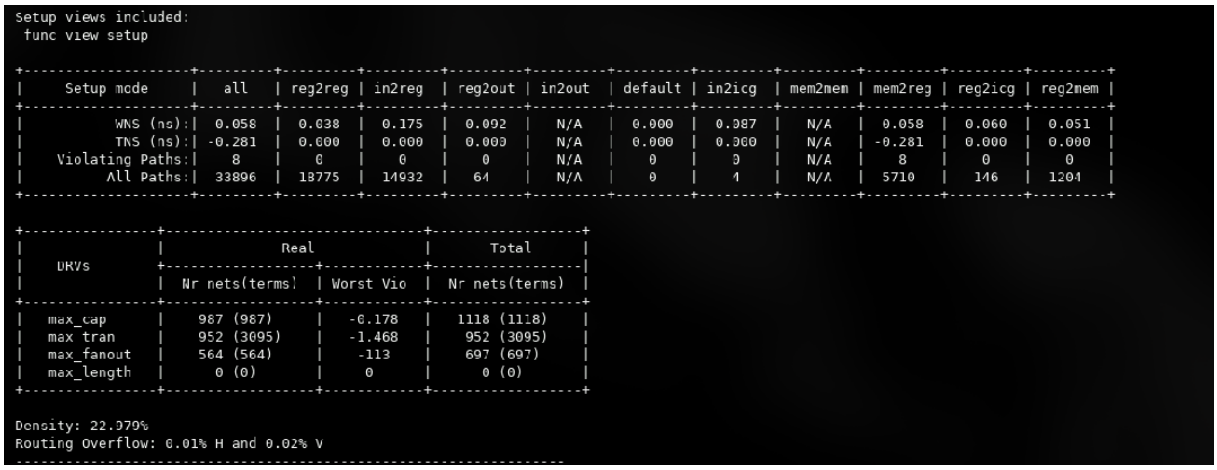


Figure 23: preCTS_optimize

After the floorplan refinements, the WNS improved to -0.058 ns with eight violating paths.

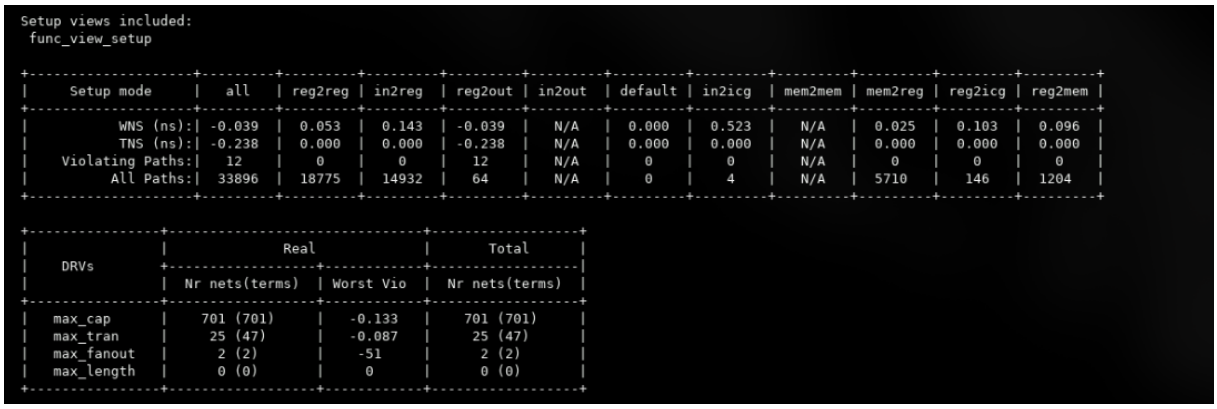


Figure 24: postCTS_v3

The CTS results were better, but post-route timing remained challenging.

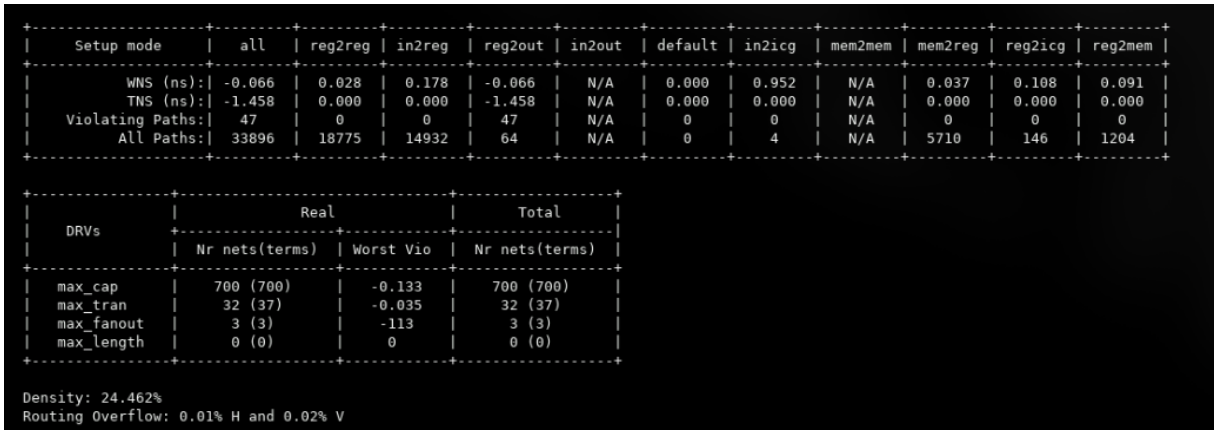


Figure 25: postRoute_v3

Further iterations were required to resolve the remaining routing-stage issues.

3.5. Comparison of Results

The different implementation experiments showed that physical design quality was strongly dependent on the interaction between floorplanning, SRAM placement, I/O placement, and physical constraints. Each iteration was therefore used not only to improve the current result, but also to identify the limitations of the previous implementation.

Experiment 1 was mainly used as a baseline. Although the design had sufficient space for standard cells and the initial power distribution network could be constructed successfully, the SRAM orientation and placement resulted in poor physical organization. The long distances between some memories and the related logic also led to poor timing performance. As a result, the 250 MHz target could not be achieved.

Experiment 2 introduced a significantly improved SRAM arrangement and provided more space for standard-cell placement. The SRAM-related DRC problems observed in the first experiment were eliminated, while I/O pins were also manually positioned instead of relying entirely on automatic assignment. After placement optimization, the reg-to-reg WNS reached approximately -0.005 ns with 51 violating paths, indicating that the 250 MHz target was becoming realistic.

After CTS, reg-to-reg timing was successfully closed through clock-tree optimization and useful skew. However, a difficult reg-to-out violation remained. Although Innovus reported this path group as *reg2out*, the actual path originated from an SRAM output. This type of memory-to-output path was significantly more difficult to improve using conventional standard-cell optimization, and the design became increasingly difficult to optimize during routing.

Experiment 3 focused more strongly on the physical relationship between the architectural blocks. The instruction memory was moved closer to the center of the design, I/O pins were repositioned toward the top of the die, and additional placement constraints were introduced for important functional blocks. These changes reduced the physical distance of critical connections and improved the overall placement structure.

The timing results improved substantially after these changes. At the pre-CTS stage, the WNS reached approximately 0.058 ns with only 8 violating paths according to the reported result. However, the remaining routing-stage timing violations showed that achieving good pre-CTS timing did not necessarily guarantee successful post-route timing closure. This became an important observation during the project: optimization at one stage could expose new problems at a later stage.

Overall, the experiments demonstrated a clear progression from a basic floorplan toward a more constraint-aware physical implementation. The main improvements did not come from a single optimization command, but from repeatedly analyzing the physical implementation, identifying the dominant problem, modifying the corresponding constraint or placement strategy, and evaluating the result again.

Table 2: Comparison of Physical Design Experiments

Exp.	Main Change	Observed Improvement	Main Limitation
1	Initial floorplan and PG construction	Established the baseline implementation and verified the initial power distribution network.	Poor SRAM organization and long physical connections resulted in poor timing; 250 MHz was not achieved.
2	Improved SRAM arrangement and manual I/O placement	SRAM DRC issues were eliminated. Pre-CTS reg-to-reg WNS reached approximately −0.005 ns with 51 violating paths.	A difficult memory-to-output timing violation remained after CTS, limiting further optimization during routing.
3	Improved architectural placement, I/O relocation, and block-level placement constraints	Pre-CTS timing improved significantly, reaching approximately 0.058 ns WNS with 8 violating paths according to the reported result.	Post-route timing remained challenging, requiring further optimization and another implementation iteration.

4. Physical Design Final Case

4.1. Final Implementation Configuration

The final implementation was obtained after multiple floorplan, placement, clock-tree, and routing iterations. The following configuration represents the main implementation constraints and optimization settings used in the final case that successfully targeted 300 MHz.

Table 3: Final Implementation Constraints

Parameter	Final Setting
Bottom routing layer	Metal 2
Top routing layer	Metal 10
Maximum transition	0.5
Setup target slack	0.020 ns
Hold target slack	0.001 ns
RC best-case temperature	0°C
RC worst-case temperature	125°C
Maximum route length	150

Table 4: Final Clock Tree and Optimization Configuration

Parameter	Final Setting
Clock buffer cells	`CLKBUF_X1, CLKBUF_X2, CLKBUF_X3`
General buffer cells	`BUF_X1, BUF_X2, BUF_X4, BUF_X8, BUF_X16, BUF_X32`
Inverter cells	`INV_X1, INV_X2, INV_X4, INV_X8, INV_X16, INV_X32`
Hold-fix cells	`BUF_X1, BUF_X2, BUF_X4, BUF_X8, BUF_X16, INV_X1, INV_X2, INV_X4`
Clock trunk top layer	Metal 8
Clock trunk bottom layer	Metal 3
Clock leaf top layer	Metal 6
Clock leaf bottom layer	Metal 2
Clock NDR layer range	Metal 2–Metal 8
Target clock skew	0.2 ns
Target clock transition	0.2 ns
Target clock fanout	24
Target clock capacitance	0.2

These settings were not selected as a single initial configuration. They were refined through repeated implementation experiments, particularly during timing optimization and clock-tree synthesis. The final configuration provided a stable basis for achieving the 300 MHz target while maintaining acceptable transition, capacitance, and clock-tree constraints.

4.2. Floorplan

After reviewing the initial placement results, I refined the floorplan to improve block organization and utilization. The updated layout groups the SRAM macros into a compact arrangement while preserving a large contiguous region for standard cells and routing resources. The Innovus screenshot in Figure 26 shows the post-placement layout alongside the utilization report from this iteration.

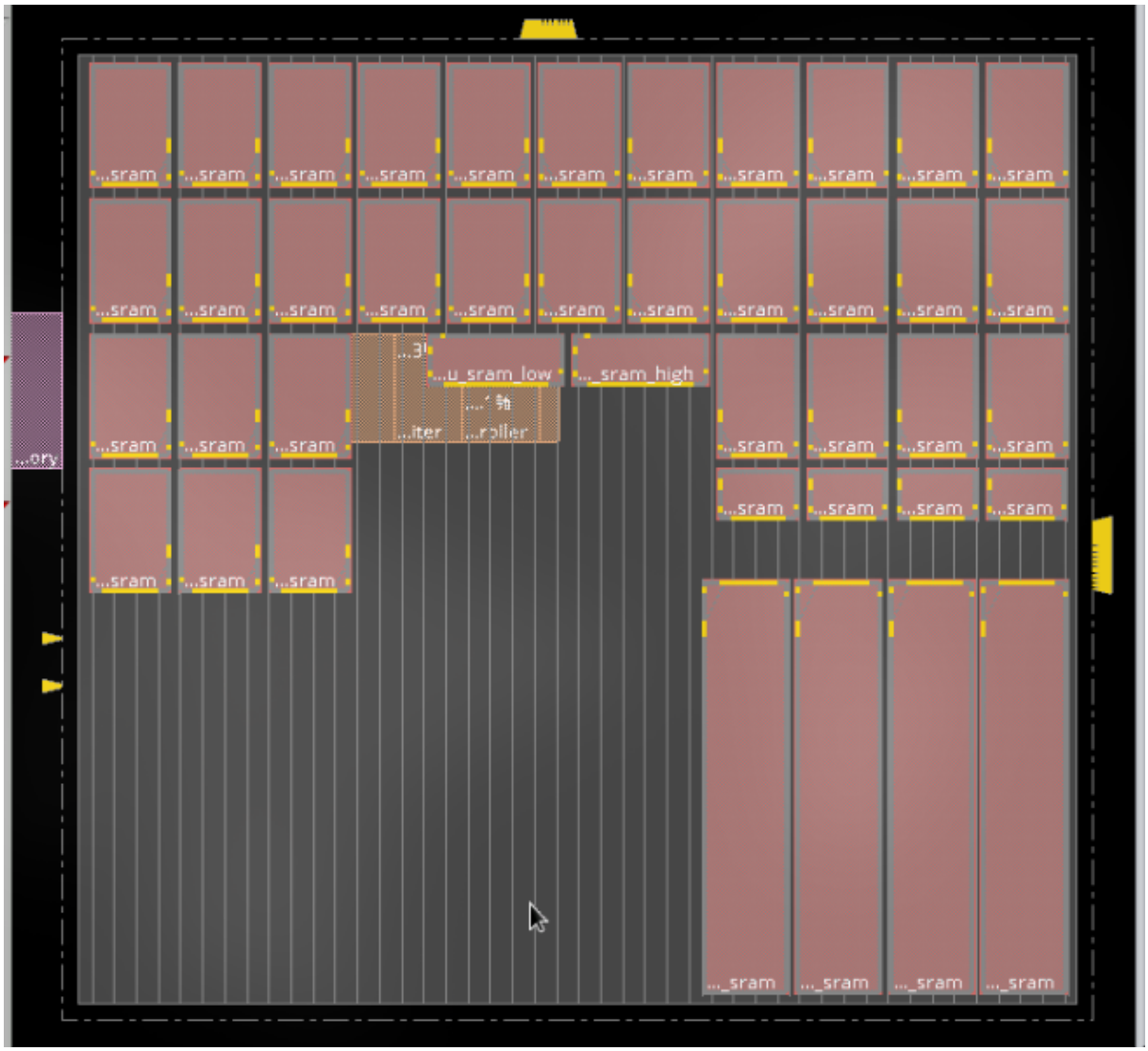


Figure 26: Floorplan

```
Checking Preroutes.....
No. of regular pre-routes not on tracks : 0

Reporting Utilizations.....

Core utilization  = 64.009678
TU for constraint module  u_global_arbiter = 11.064163
TU for constraint module  u_controller = 47.078946
TU for constraint module  u_ifm_arbiter = 20.324858
TU for constraint module  u_line_buffer = 35.246573
TU for constraint module  u_pea = 122143369600000.000000
Effective Utilizations

Average macro density = 0.000.
Average module density = 0.234.
Density for module 'u_global_arbiter' = 0.227.
       = stdcell_area 1414 sites (376 um^2) / alloc_area 6241 sites (1660 um^2).
Density for module 'u_controller' = 0.476.
       = stdcell_area 24635 sites (6553 um^2) / alloc_area 51759 sites (13768 um^2).
Density for module 'u_ifm_arbiter' = 0.273.
       = stdcell_area 18194 sites (4840 um^2) / alloc_area 66582 sites (17711 um^2).
Density for module 'u_line_buffer' = 0.356.
       = stdcell_area 20234 sites (5382 um^2) / alloc_area 56851 sites (15122 um^2).
Pure std cell Density 0.230934
 Density for the rest of the design = 0.227.
       = stdcell_area 1172379 sites (311853 um^2) / alloc_area 5174447 sites (1376403 um^2).
Density for the design = 0.231.
       = stdcell_area 1236856 sites (329004 um^2) / alloc_area 5355880 sites (1424664 um^2).
Pin Density = 0.06586.
            = total # of pins 762052 / total area 11570196.
```

Figure 27: Floorplan utilization

The reported core utilization is approximately 64%, which is moderate and helps keep the design routable without excessive congestion. The implementation session also confirmed that the floorplan did not introduce any pre-route track issues, which made this case the strongest candidate for the subsequent placement, CTS, and routing steps.

4.3. Placement Optimization

```
Setup views included:
 func_view_setup

+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | in2reg  | reg2out | in2out  | default | in2icg  | mem2mem | mem2reg | reg2icg | reg2mem |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):| -0.033  |  0.038  |  0.101  |  0.073  |   N/A   |  0.000  |  0.202  |   N/A   | -0.033  |  0.072  |  0.053  |
|           TNS (ns):| -0.113  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |   N/A   | -0.113  |  0.000  |  0.000  |
|    Violating Paths:|    6    |    0    |    0    |    0    |   N/A   |    0    |    0    |   N/A   |    6    |    0    |    0    |
|          All Paths:|  33896  |  18775  |  14932  |   64    |   N/A   |    0    |    1    |   N/A   |  5710   |   146   |  1204   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+

+---------------+-----------------------------+-----------------+
|               |             Real            |       Total     |
|    DRVs       +-----------------+-----------+-----------------|
|               |  Nr nets(terms) | Worst Vio |  Nr nets(terms) |
+               +                 +           +                 +
|   max_cap     |   1019 (1019)   |   -0.128  |   1151 (1151)   |
|   max_tran    |   1031 (3762)   |   -1.552  |   1031 (3762)   |
|   max_fanout  |    540 (540)    |    -113   |    673 (673)    |
|   max_length  |      0 (0)      |     0     |      0 (0)      |
+---------------+-----------------+-----------+-----------------+

Density: 23.093%
Routing Overflow: 0.01% H and 0.01% V
```

Figure 28: preCTS timing report

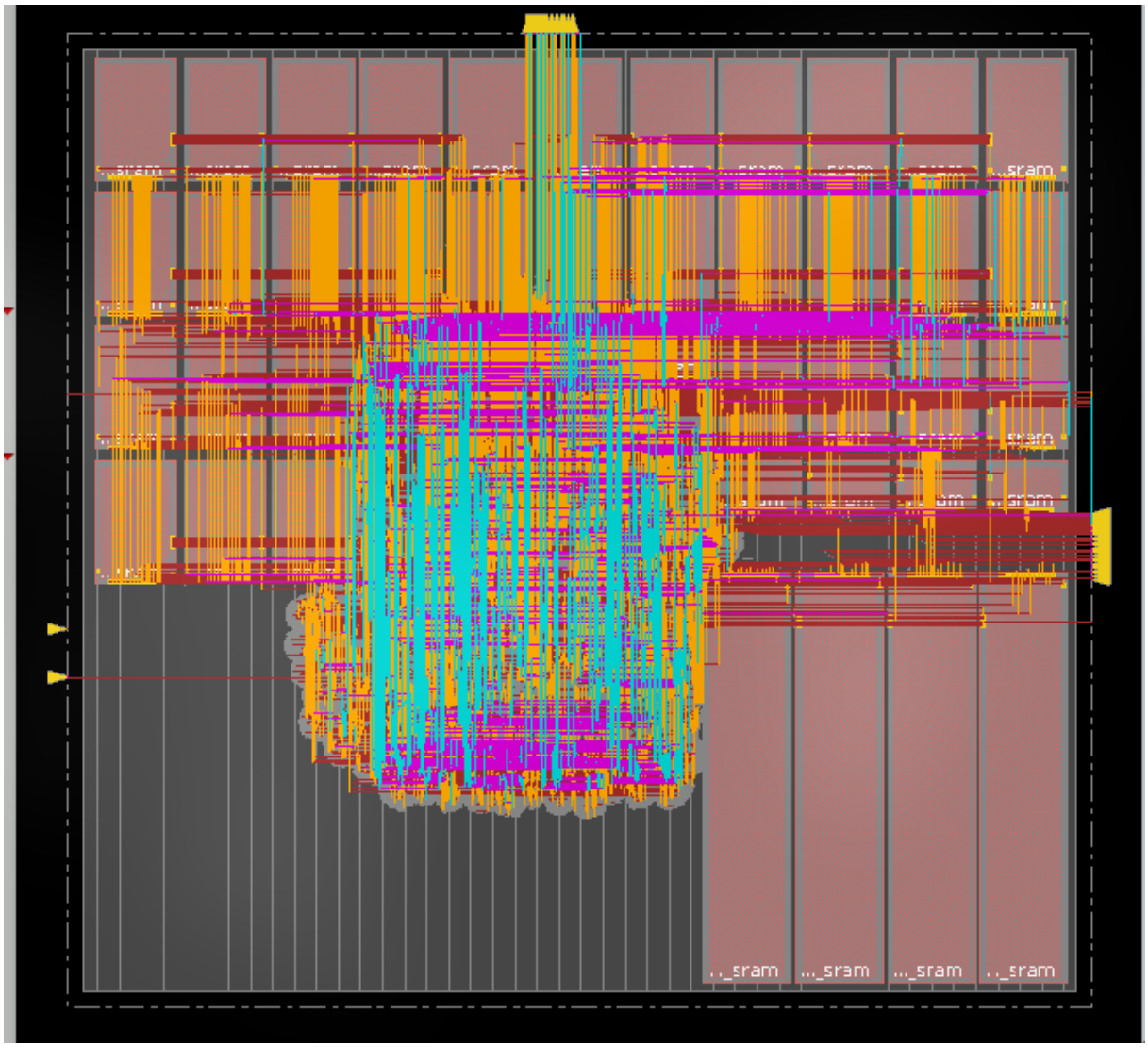
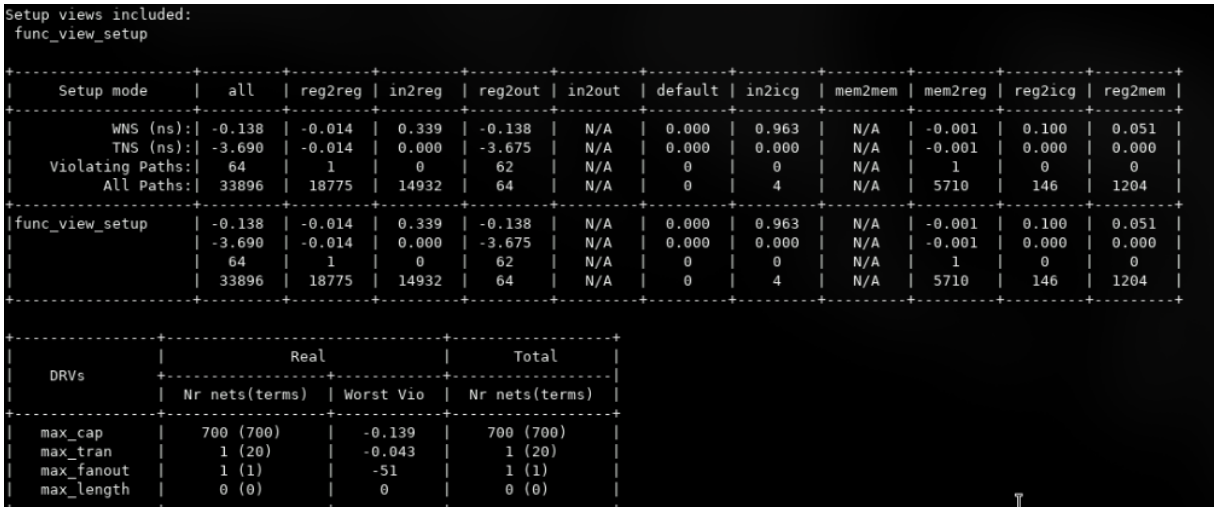


Figure 29: Overview of early routing

These results confirm the improved placement quality.

4.4. Clock Tree Synthesis Optimization



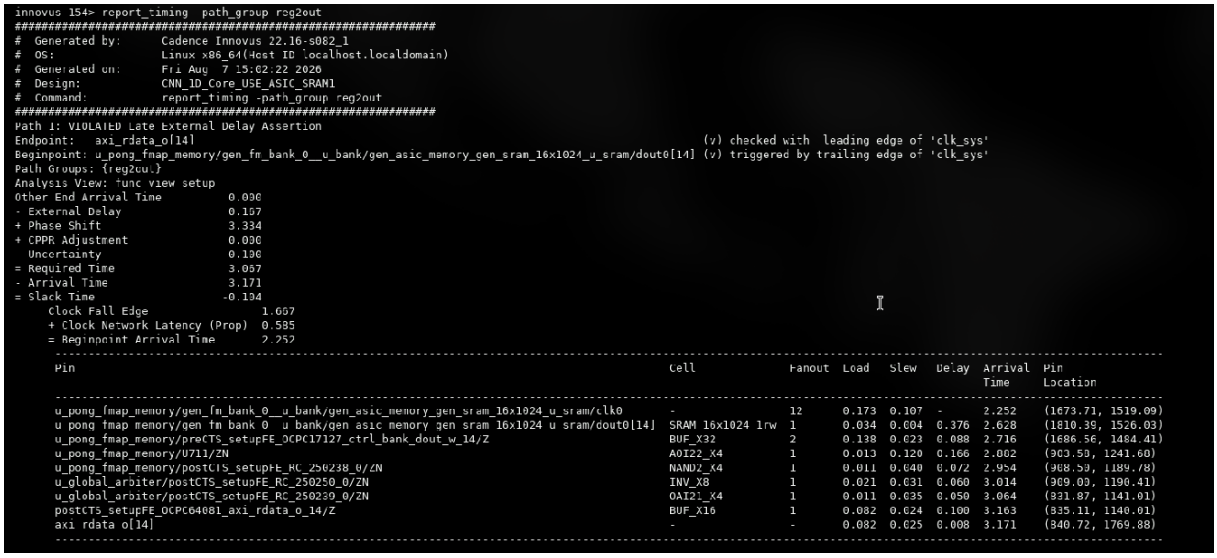
```
Setup views included:
 func_view_setup

+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | in2reg  | reg2out | in2out  | default | in2icg  | mem2mem | mem2reg | reg2icg | reg2mem |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):| -0.138  | -0.014  |  0.339  | -0.138  |   N/A   |  0.000  |  0.963  |   N/A   | -0.001  |  0.100  |  0.051  |
|           TNS (ns):| -3.690  | -0.014  |  0.000  | -3.675  |   N/A   |  0.000  |  0.000  |   N/A   | -0.001  |  0.000  |  0.000  |
|    Violating Paths:|   64    |    1    |    0    |   62    |   N/A   |    0    |    0    |   N/A   |    1    |    0    |    0    |
|          All Paths:|  33896  |  18775  |  14932  |   64    |   N/A   |    0    |    4    |   N/A   |  5710   |   146   |  1204   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|func_view_setup     | -0.138  | -0.014  |  0.339  | -0.138  |   N/A   |  0.000  |  0.963  |   N/A   | -0.001  |  0.100  |  0.051  |
|                    | -3.690  | -0.014  |  0.000  | -3.675  |   N/A   |  0.000  |  0.000  |   N/A   | -0.001  |  0.000  |  0.000  |
|                    |   64    |    1    |    0    |   62    |   N/A   |    0    |    0    |   N/A   |    1    |    0    |    0    |
|                    |  33896  |  18775  |  14932  |   64    |   N/A   |    0    |    4    |   N/A   |  5710   |   146   |  1204   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+


+----------------+-------------------------------+------------------+
|                |             Real              |       Total      |
|    DRVs        +-------------------+-----------+------------------+
|                |  Nr nets(terms)   | Worst Vio |  Nr nets(terms)  |
+----------------+-------------------+-----------+------------------+
|   max_cap      |    700 (700)      |   -0.139  |    700 (700)     |
|   max_tran     |      1 (20)       |   -0.043  |      1 (20)      |
|   max_fanout   |      1 (1)        |    -51    |      1 (1)       |
|   max_length   |      0 (0)        |     0     |      0 (0)       |
```

Figure 30: postCTS timing report

After CTS, the ‘mem2out‘ path remained the worst timing violation.



```
innovus 154> report_timing -path_group reg2out
#################################################################
#  Generated by:      Cadence Innovus 22.16-s082_1
#  OS:                Linux x86_64(Host ID localhost.localdomain)
#  Generated on:      Fri Aug  7 15:02:22 2026
#  Design:            CNN_1D_Core_USE_ASIC_SRAM1
#  Command:           report_timing -path_group reg2out
#################################################################
Path 1: VIOLATED Late External Delay Assertion
Endpoint:   axi_rdata_o[14]                                                    (v) checked with  leading edge of 'clk_sys'
Beginpoint: u_pong_fmap_memory/gen_fm_bank_0__u_bank/gen_asic_memory_gen_sram_16x1024_u_sram/dout0[14] (v) triggered by trailing edge of 'clk_sys'
Path Groups: {reg2out}
Analysis View: func_view_setup
Other End Arrival Time          0.000
- External Delay                0.167
+ Phase Shift                   3.334
+ CPPR Adjustment               0.000
  Uncertainty                   0.130
= Required Time                 3.037
- Arrival Time                  3.171
= Slack Time                   -0.134
     Clock Fall Edge             1.667
     + Clock Network Latency (Prop)  0.585
     = Beginpoint Arrival Time       2.252
     ------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
     Pin                                                                Cell              Fanout  Load   Slew   Delay  Arrival  Pin
                                                                                                                          Time     Location
     ------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
     u_pong_fmap_memory/gen_fm_bank_0__u_bank/gen_asic_memory_gen_sram_16x1024_u_sram/clk0                -                 12      0.173  0.107  -      2.252    (1673.71, 1519.09)
     u_pong_fmap_memory/gen_fm_bank_0__u_bank/gen_asic_memory_gen_sram_16x1024_u_sram/dout0[14]  SRAM_16x1024_1rw   1       0.034  0.004  0.376  2.628    (1810.39, 1526.03)
     u_pong_fmap_memory/preCTS_setupFE_OCPC17127_ctrl_bank_dout_w_14/Z                  BUF_X32           2       0.138  0.023  0.088  2.716    (1686.56, 1484.41)
     u_pong_fmap_memory/U711/ZN                                         AOI22_X4          1       0.013  0.120  0.166  2.882    (963.53, 1241.60)
     u_pong_fmap_memory/postCTS_setupFE_RC_250238_0/ZN                  NAND2_X4          1       0.011  0.040  0.072  2.954    (968.53, 1189.78)
     u_global_arbiter/postCTS_setupFE_RC_250250_0/ZN                    INV_X8            1       0.021  0.031  0.060  3.014    (969.03, 1190.41)
     u_global_arbiter/postCTS_setupFE_RC_250239_0/ZN                    OAI21_X4          1       0.011  0.035  0.050  3.064    (831.87, 1141.01)
     postCTS_setupFE_OCPC64081_axi_rdata_o_14/Z                         BUF_X16           1       0.082  0.024  0.100  3.163    (835.11, 1140.01)
     axi_rdata_o[14]                                                    -                 -       0.082  0.025  0.008  3.171    (840.72, 1769.88)
     ------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
```

Figure 31: mem2out path report

The path report indicates minimal logic along the critical path and an already optimized slew and delay profile.

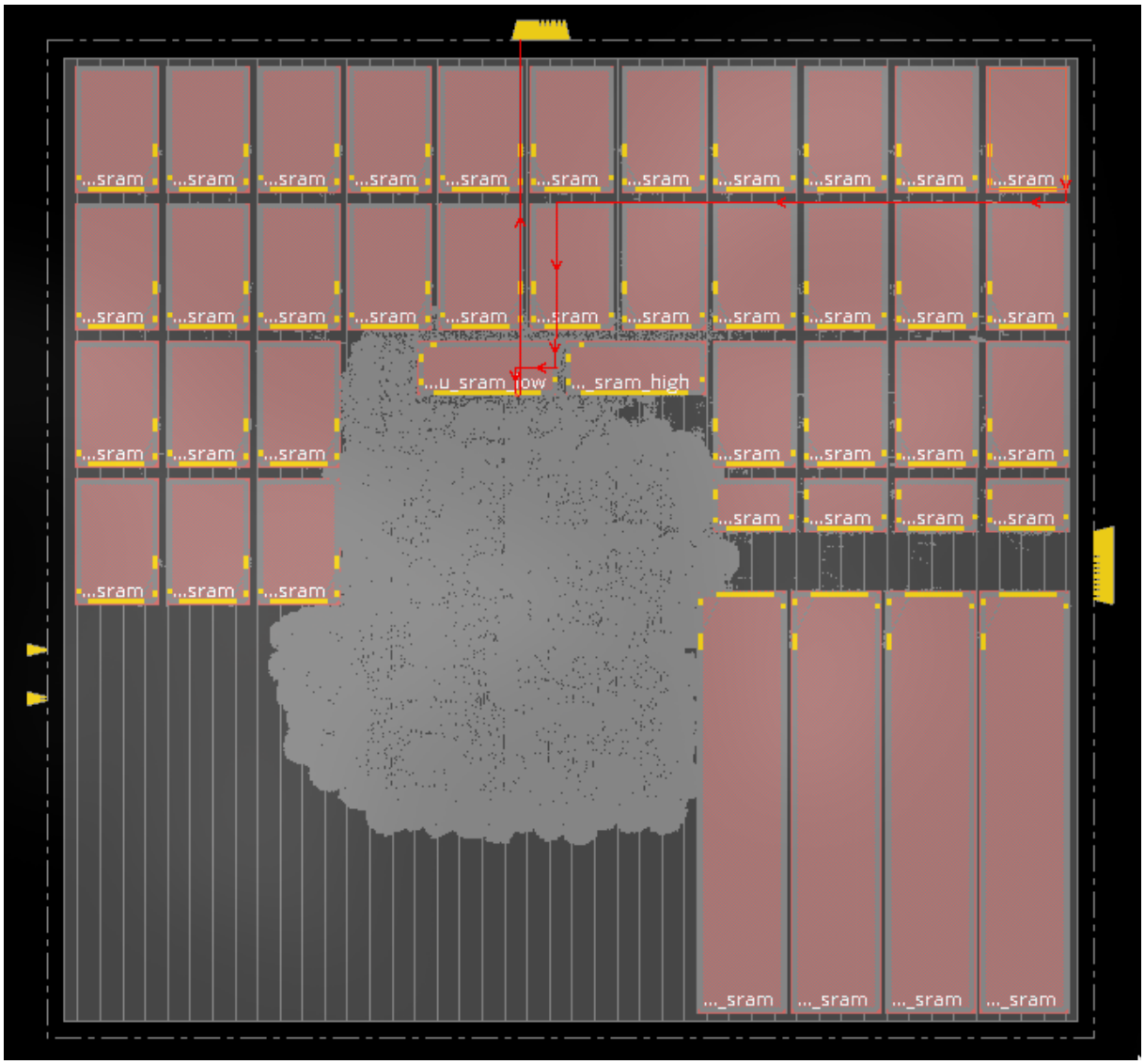


Figure 32: display worst path mem2out

The path passes through ‘u_instruction_memory‘ before reaching the output, which appears to be the shortest available route. A detailed review showed that external delay settings in the SDC file could be reduced and that the clock network latency was high at 0.585 ns. Repositioning the clock pin was therefore necessary.

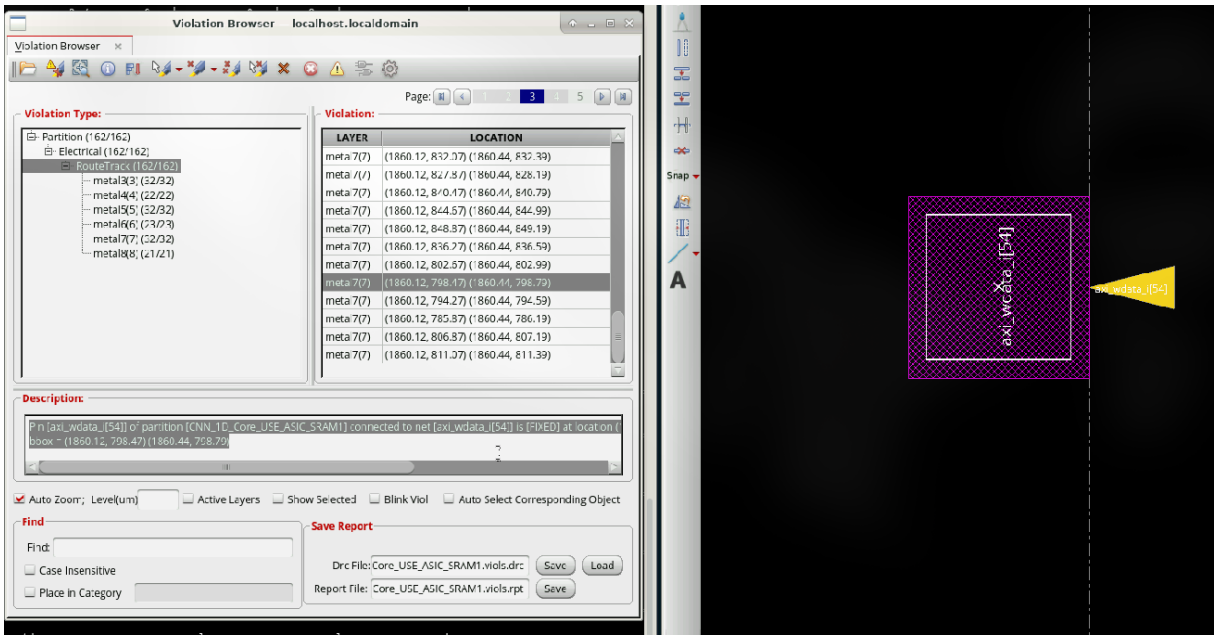


Figure 33: checkPinAssignment

The clock pin assignment was adjusted to align better with the routing tracks.

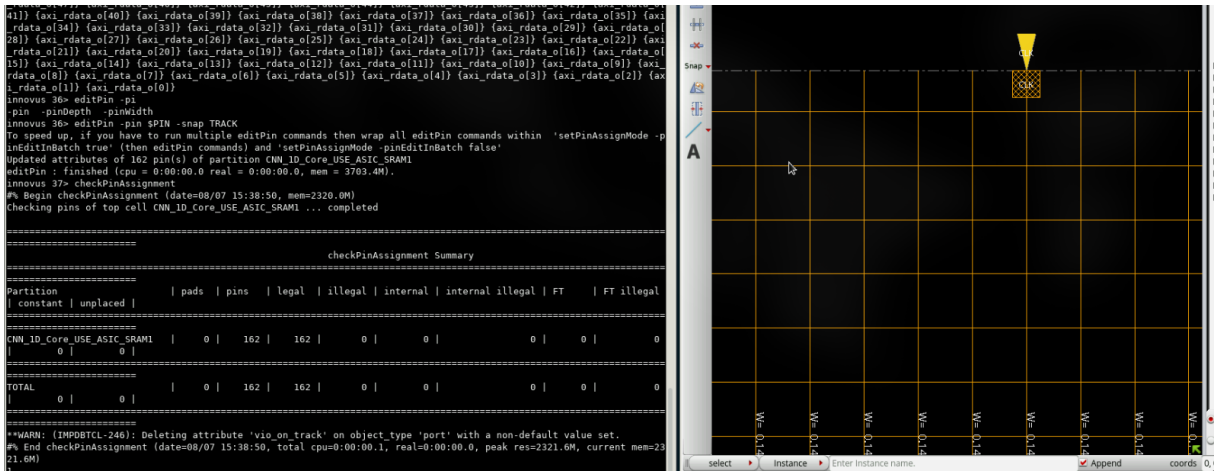


Figure 34: Clock check

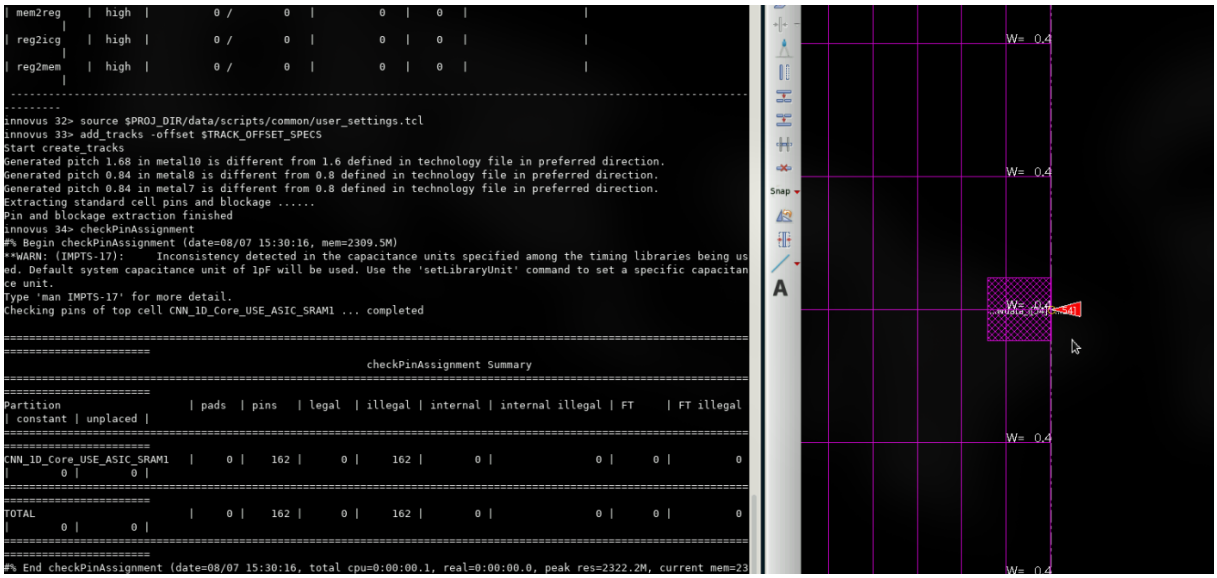


Figure 35: Random check I/O pin

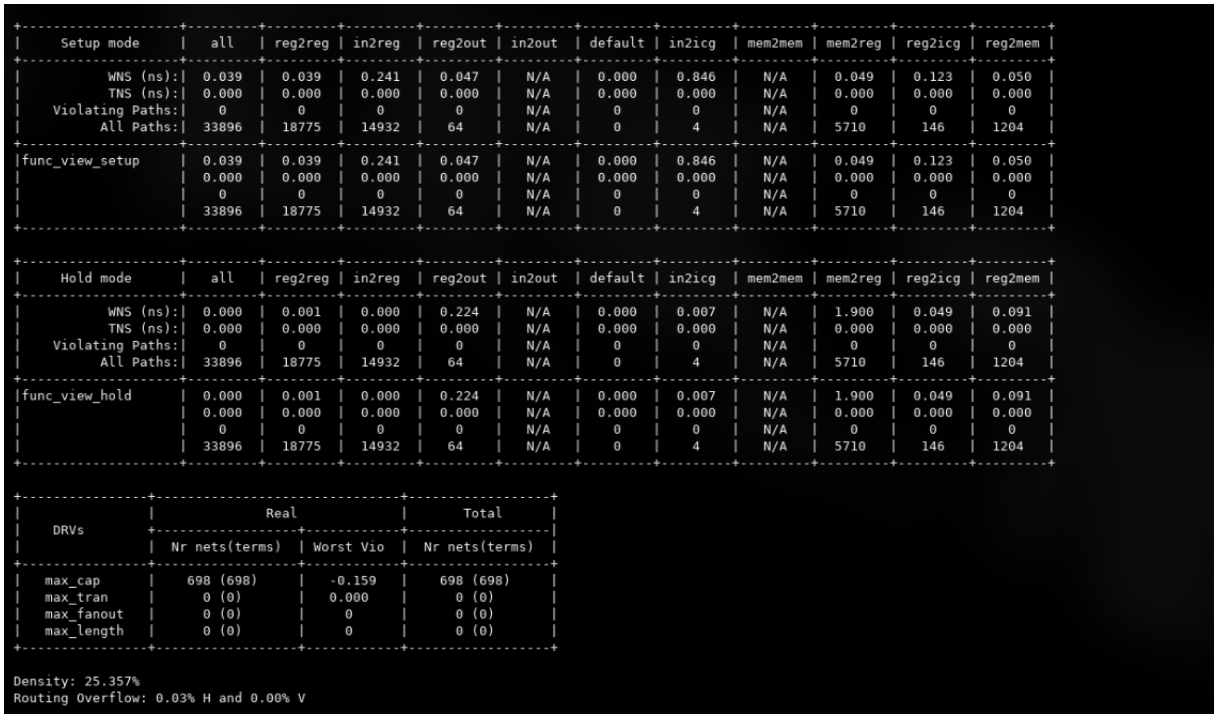


Figure 36: timing report after replace_clk_pin

This targeted CTS adjustment improved timing and provided a stronger basis for routing.

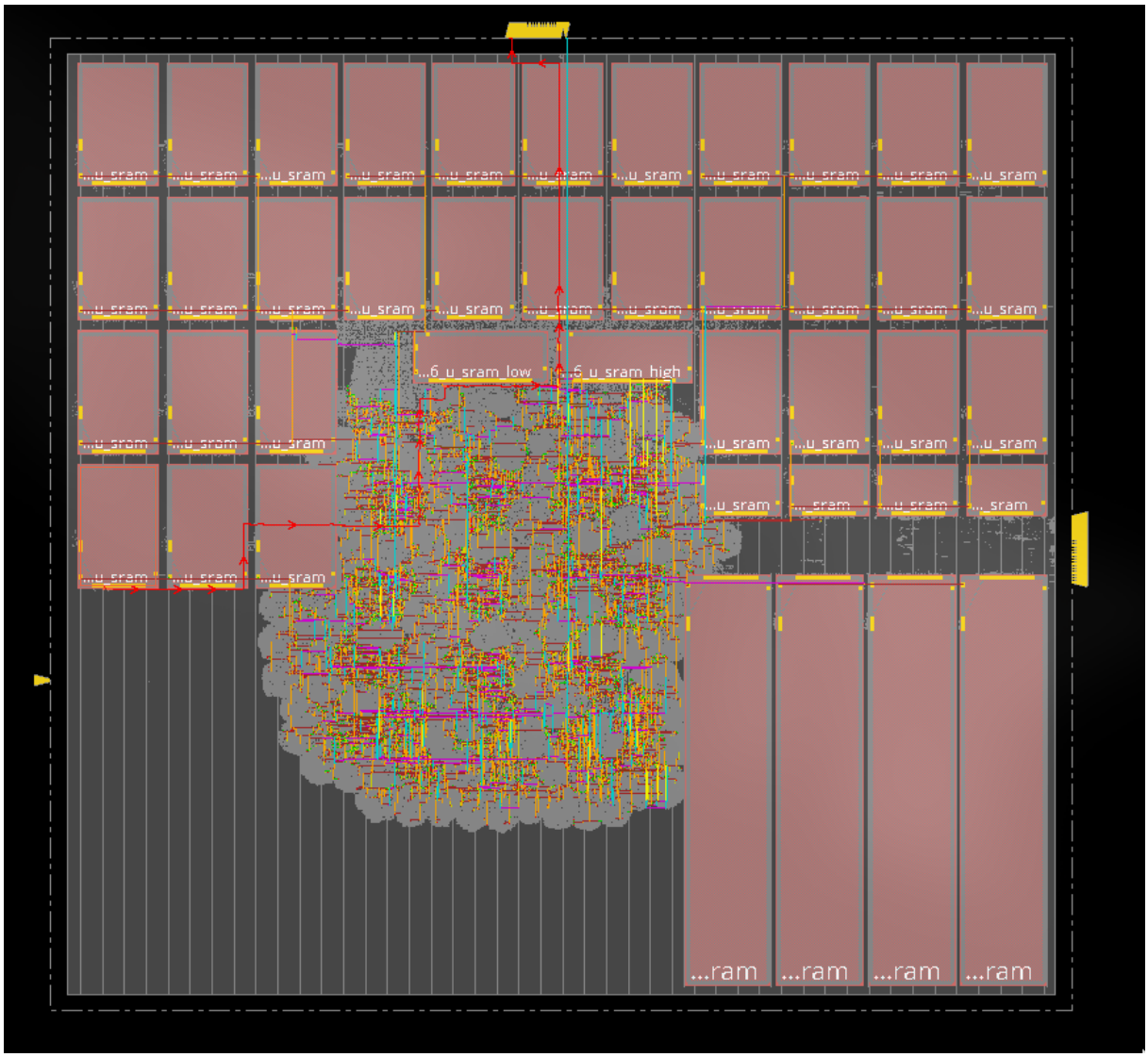


Figure 37: Final Clock Tree

4.5. Routing Optimization & Final Inspection Steps

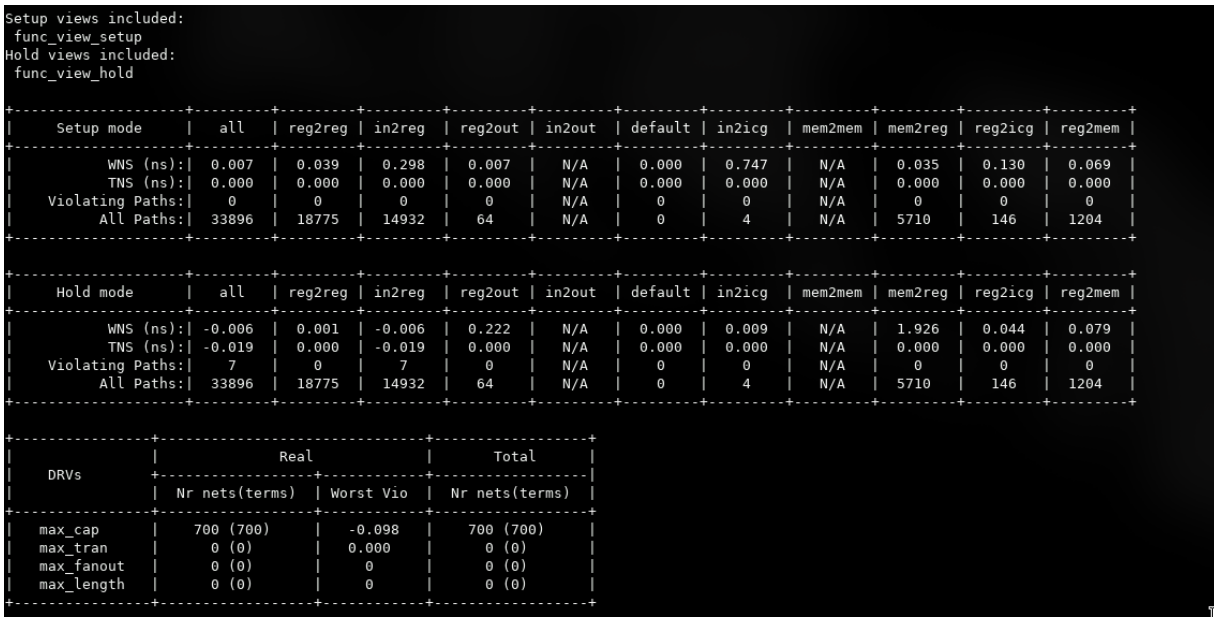
Setup views included:
 func_view_setup

Setup mode	all	reg2reg	in2reg	reg2out	in2out	default	in2icg	mem2mem	mem2reg	reg2icg	reg2mem
WNS (ns):	0.007	0.039	0.350	0.007	N/A	0.000	0.746	N/A	0.035	0.130	0.069
TNS (ns):	0.000	0.000	0.000	0.000	N/A	0.000	0.000	N/A	0.000	0.000	0.000
Violating Paths:	0	0	0	0	N/A	0	0	N/A	0	0	0
All Paths:	33896	18775	14932	64	N/A	0	4	N/A	5710	146	1204

DRVs	Real		Total	
	Nr nets(terms)	Worst Vio	Nr nets(terms)	
max_cap	700 (700)	-0.098	700 (700)	
max_tran	0 (0)	0.000	0 (0)	
max_fanout	0 (0)	0	0 (0)	
max_length	0 (0)	0	0 (0)	

Figure 38: postRoute

After routing optimization, setup timing was met and hold timing was then verified.



```
Setup views included:
 func_view_setup
Hold views included:
 func_view_hold

+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | in2reg  | reg2out | in2out  | default | in2icg  | mem2mem | mem2reg | reg2icg | reg2mem |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.007  |  0.039  |  0.298  |  0.007  |   N/A   |  0.000  |  0.747  |   N/A   |  0.035  |  0.130  |  0.069  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |   N/A   |    0    |    0    |   N/A   |    0    |    0    |    0    |
|          All Paths:|  33896  |  18775  |  14932  |   64    |   N/A   |    0    |    4    |   N/A   |  5710   |   146   |  1204   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+

+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | in2reg  | reg2out | in2out  | default | in2icg  | mem2mem | mem2reg | reg2icg | reg2mem |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):| -0.006  |  0.001  | -0.006  |  0.222  |   N/A   |  0.000  |  0.009  |   N/A   |  1.926  |  0.044  |  0.079  |
|           TNS (ns):| -0.019  |  0.000  | -0.019  |  0.000  |   N/A   |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    7    |    0    |    7    |    0    |   N/A   |    0    |    0    |   N/A   |    0    |    0    |    0    |
|          All Paths:|  33896  |  18775  |  14932  |   64    |   N/A   |    0    |    4    |   N/A   |  5710   |   146   |  1204   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |             Real              |       Total      |
|    DRVs        +-----------------+-------------+------------------|
|                |  Nr nets(terms) | Worst Vio   |  Nr nets(terms)  |
+----------------+-----------------+-------------+------------------+
|   max_cap      |    700 (700)    |   -0.098    |    700 (700)     |
|   max_tran     |      0 (0)      |   0.000     |      0 (0)       |
|   max_fanout   |      0 (0)      |     0       |      0 (0)       |
|   max_length   |      0 (0)      |     0       |      0 (0)       |
+----------------+-----------------+-------------+------------------+
```

Figure 39: postRoute



```
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | in2reg  | reg2out | in2out  | default | in2icg  | mem2mem | mem2reg | reg2icg | reg2mem |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.007  |  0.039  |  0.297  |  0.007  |   N/A   |  0.000  |  0.746  |   N/A   |  0.035  |  0.130  |  0.069  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |   N/A   |    0    |    0    |   N/A   |    0    |    0    |    0    |
|          All Paths:|  33896  |  18775  |  14932  |   64    |   N/A   |    0    |    4    |   N/A   |  5710   |   146   |  1204   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+

+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | in2reg  | reg2out | in2out  | default | in2icg  | mem2mem | mem2reg | reg2icg | reg2mem |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+
|           WNS (ns):|  0.001  |  0.001  |  0.001  |  0.222  |   N/A   |  0.000  |  0.009  |   N/A   |  1.926  |  0.044  |  0.079  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |   N/A   |  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |    0    |   N/A   |    0    |    0    |   N/A   |    0    |    0    |    0    |
|          All Paths:|  33896  |  18775  |  14932  |   64    |   N/A   |    0    |    4    |   N/A   |  5710   |   146   |  1204   |
+--------------------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |             Real              |       Total      |
|    DRVs        +-----------------+-------------+------------------|
|                |  Nr nets(terms) | Worst Vio   |  Nr nets(terms)  |
+----------------+-----------------+-------------+------------------+
|   max_cap      |    700 (700)    |   -0.098    |    700 (700)     |
|   max_tran     |      0 (0)      |   0.000     |      0 (0)       |
|   max_fanout   |      0 (0)      |     0       |      0 (0)       |
|   max_length   |      0 (0)      |     0       |      0 (0)       |
+----------------+-----------------+-------------+------------------+

Density: 25.756%
```

Figure 40: postRoute_opt

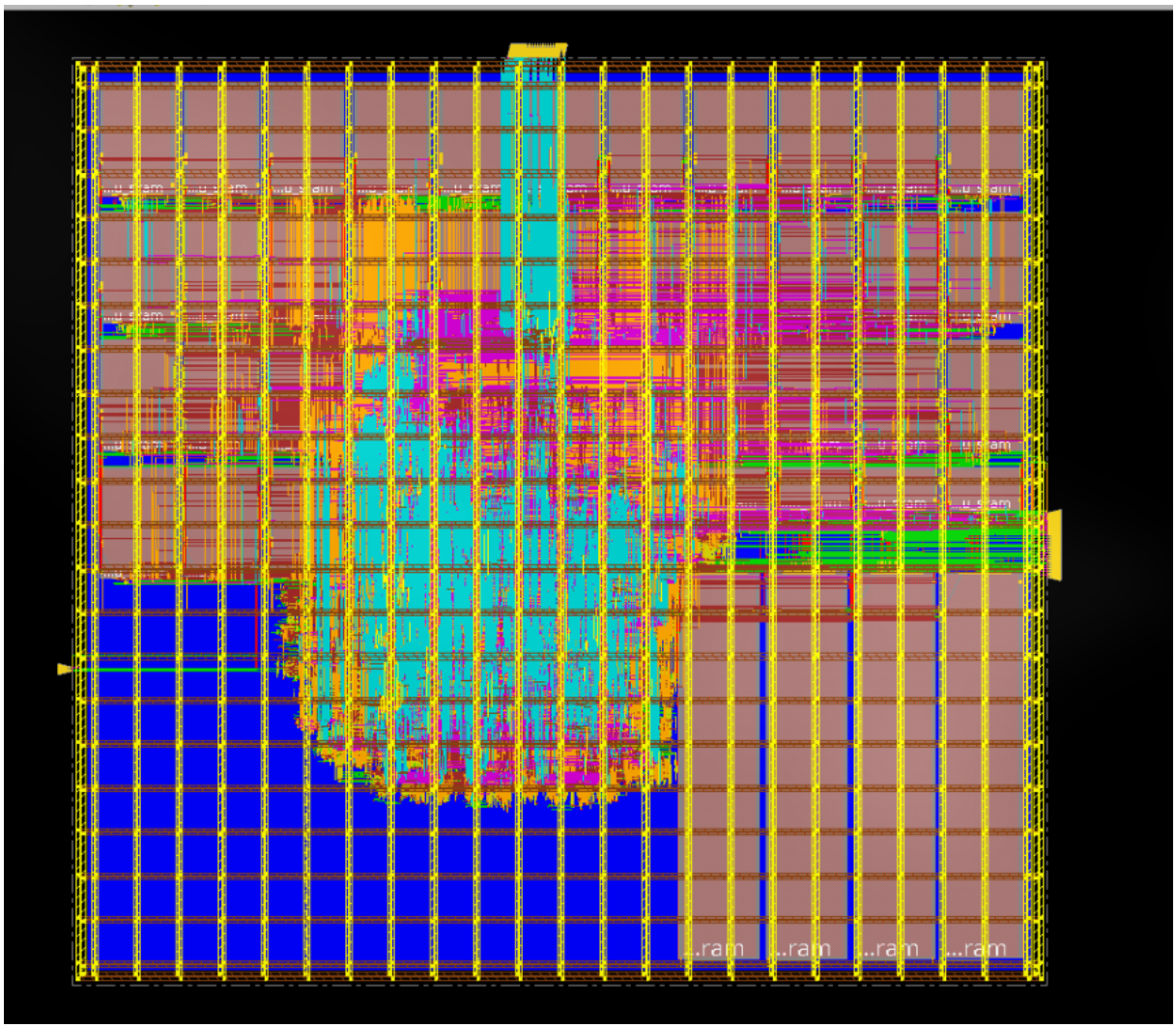


Figure 41: Overall design overview

Setup and hold timing were clean, and the implementation moved on to connectivity and DRC verification.

```
******** Start: VERIFY CONNECTIVITY ********
Start Time: Sat Aug  8 01:02:41 2026

Design Name: CNN_1D_Core_USE_ASIC_SRAM1
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (1860.4800, 1769.8800)
Error Limit = 1000; Warning Limit = 50
Check specified nets
Use 32 pthreads

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Sat Aug  8 01:02:45 2026
Time Elapsed: 0:00:04.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:23.3  MEM: 0.000M)
```

Figure 42: Connectivity report

No connection errors were found.

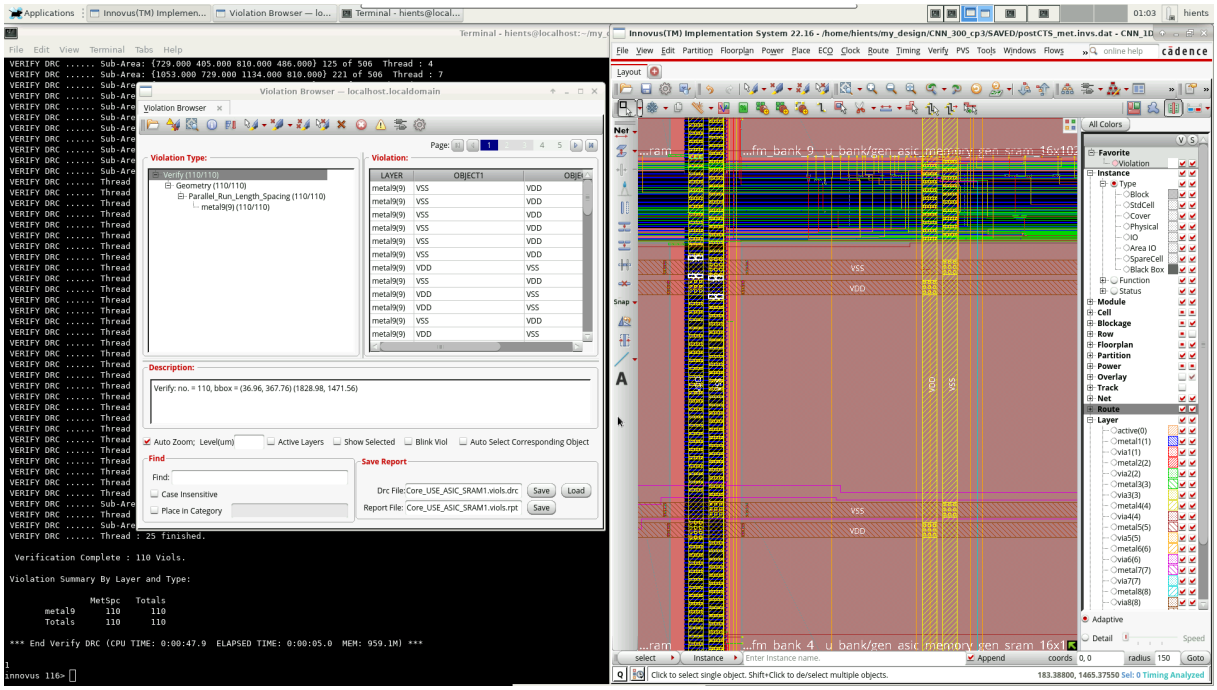


Figure 43: DRC report

No shorts, opens, or antenna issues were detected. A few parallel run-length spacing warnings remained on Metal 9 power lines, and these were accepted at this stage.

Table 5: Final Power Breakdown

Group	Internal (mW)	Switching (mW)	Leakage (mW)	Total (mW)
Sequential	19.31	1.919	1.041	22.27
Macro	55.35	0.025	0.851	56.22
Combinational	29.95	29.42	2.984	62.35
Clock	0.225	0	0.006	0.231
Total	**104.82**	**31.36**	**4.88**	**141.07**

The final implementation consumes approximately 141.07 mW at a supply voltage of 0.95 V. Internal power is the dominant component, accounting for 74.31% of the total power, while switching and leakage power contribute 22.23% and 3.46%, respectively.

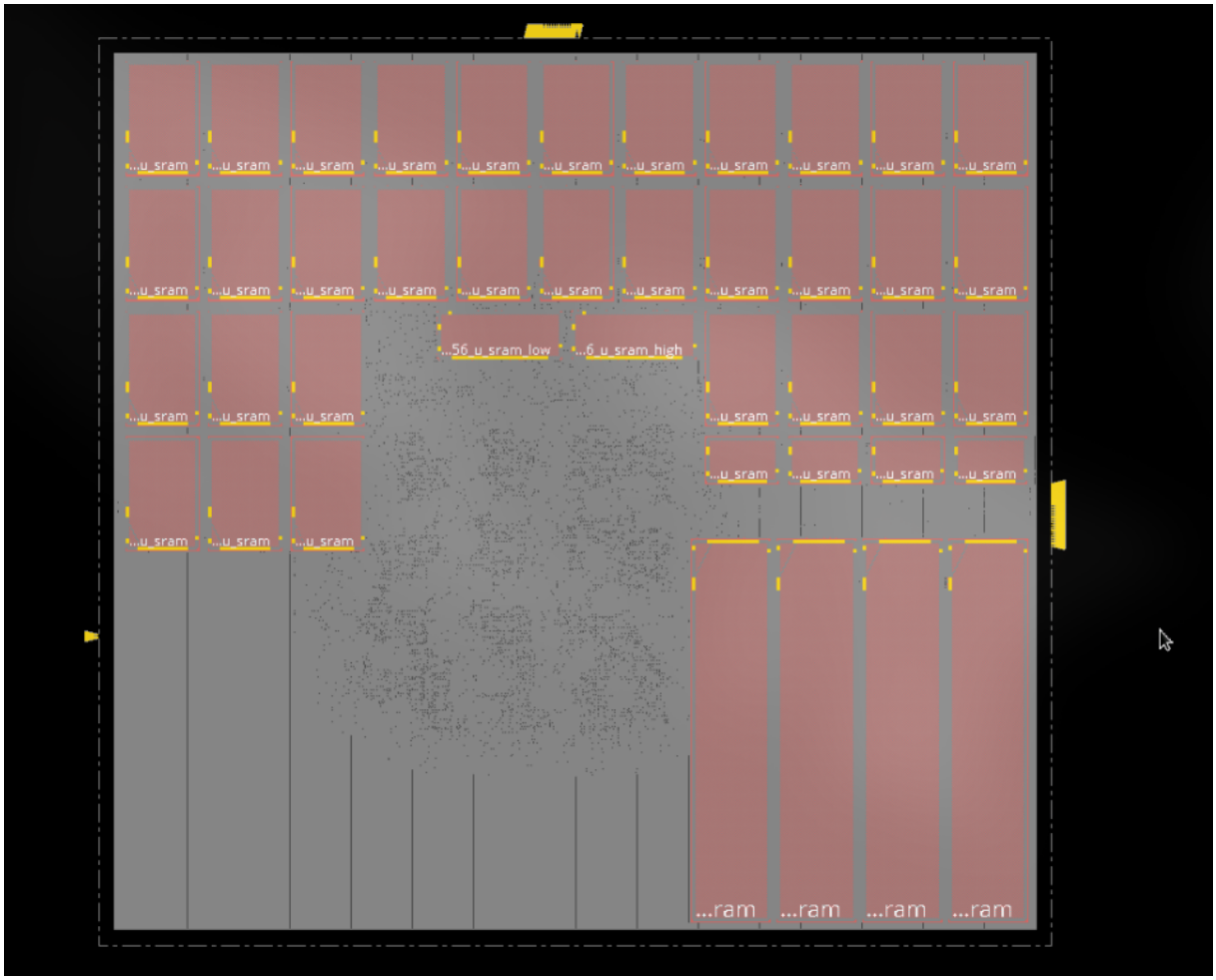


Figure 44: Add FILLER cells

FILLER cells were added to ensure physical completion and maintain uniform density across the die. These cells also support manufacturability and consistent electrical performance.

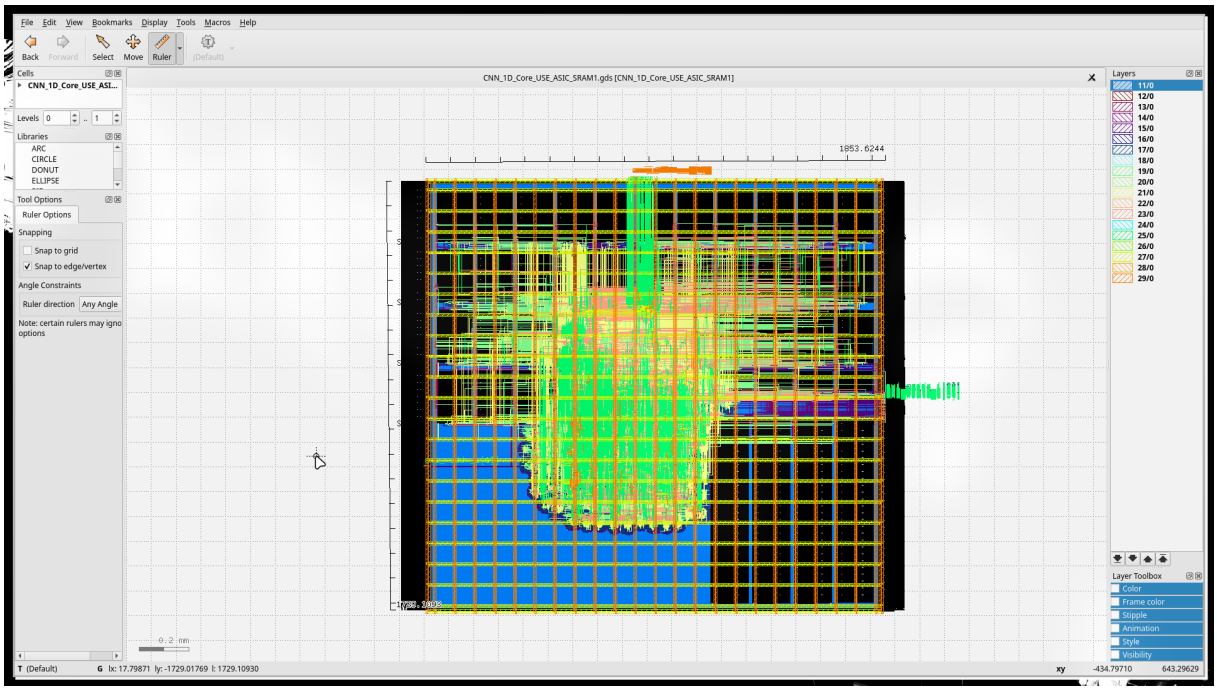


Figure 45: GDSII review

5. Debugging and Problem Solving

5.1. Root-Cause Analysis

The primary root cause of the remaining timing violations was the physical distance between the SRAM macros and the core logic, combined with an unfavorable I/O pin assignment. This resulted in long critical paths that were difficult to optimize after placement and CTS.

Other contributing factors included an incomplete clock-pin assignment and conservative external delay settings in the SDC file. These conditions amplified the impact of the ‘mem2out‘ paths and limited the effectiveness of standard timing closure techniques.

5.2. Failed Approaches

One failed approach was relying too heavily on automatic tool pin assignment during the early implementation stages. This allowed the tool to place I/O pins in locations that increased critical routing distance and clock latency.

Another less effective strategy was attempting to fix timing through only incremental placement optimization after the first two experiments. Without stronger block-level floorplanning and explicit placement constraints, the design could not recover the lost timing margin.

5.3. Successful Fixes

Two changes proved most effective. First, repositioning the ‘u_instruction_memory‘ block toward the center and moving the I/O pins to the top of the die significantly reduced critical path lengths.

Second, applying targeted block placement constraints for the core functional units improved the physical hierarchy and reduced congestion. A focused CTS adjustment to the clock pin assignment also reduced clock network latency and helped close the remaining timing violations.

5.4. Lessons from Iteration

The implementation experience highlighted that physical design is inherently iterative. Each stage can expose new issues, and early decisions in floorplanning and pin assignment have a disproportionate impact on final timing.

The project also demonstrated the value of using multiple implementation experiments in parallel. Running additional timing and placement cases provided insight into feasible targets and helped identify which improvements were most promising.

6. Results and Comparison

6.1. Initial vs. Final Results

The initial implementation served as a baseline with acceptable power distribution but poor timing due to suboptimal SRAM placement and long critical routes. The final implementation addressed these issues through improved floorplanning, a stronger placement strategy, and a more careful CTS flow.

The final design showed considerable improvement in timing structure, with a successful CTS result and a clean post-route timing profile after the last optimization steps. The final power estimate remained consistent with expectations, and the design achieved a stable implementation state.

6.2. Optimization Summary

The optimization process can be summarized as follows:

- Experiment 1: Established the initial power grid and placement baseline; revealed critical floorplan and SRAM orientation problems.
- Experiment 2: Improved SRAM arrangement and manual I/O placement; closed reg-to-reg timing but exposed difficult memory-to-output violations.
- Experiment 3: Refined architectural placement, added block constraints, and improved CTS pin assignment; delivered the best overall timing and routing behavior.

6.3. Key Improvements

The most important improvements were:

- Reorganizing SRAM placement to reduce critical path length.
- Applying manual I/O pin assignment and clock pin correction to reduce routing and clock latency.
- Adding placement constraints for the main architectural blocks to improve physical hierarchy.
- Iteratively refining the CTS and routing flow to ensure timing closure at the final stage.

6.4. Final Design Status

The final implementation achieved a stable physical design with clean setup and hold timing after the last routing optimization pass. Power and connectivity checks passed, and only minor power-grid warnings remained in noncritical Metal layers.

The design is ready for final signoff pending any remaining formal physical verification checks and GDSII validation.

7. Conclusion

7.1. Achievements

This project successfully completed the RTL-to-GDSII physical implementation flow for the CNN accelerator design. The final implementation met the targeted timing and maintained a robust physical structure through careful floorplanning, placement, and CTS optimization.

7.2. Limitations

The most significant limitation was the sensitivity of the design to SRAM placement and I/O pin assignment. These early placement decisions required multiple iterations to stabilize, and some final timing improvements were still constrained by the fixed macro locations.

7.3. Future Work

Future work should focus on further improving the I/O pin plan and exploring additional clock network optimizations. A dedicated review of the memory-to-output paths and any remaining power-grid warnings would also strengthen the final design closure.